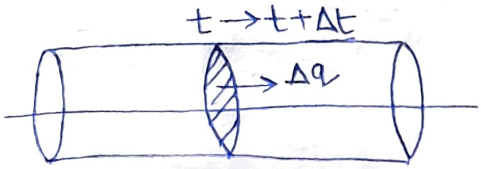
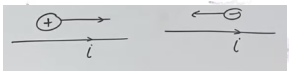
Current Electricity

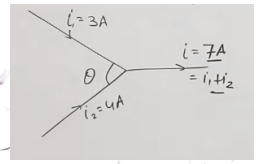
Electric Current: It is time rate of flow of charge.



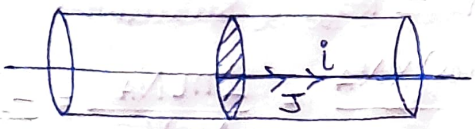
$$i_{avg} = \frac{\Delta q}{\Delta t} \qquad i_{inst} = \frac{dq}{dt}$$

→ Unit = C/sec = Ampere
→ Dimension = [I] = [A]
→ Scalar Quantity (Although it has direction, it follows scalar algebra)
→ Direction: Along movement of +ve charge
Opposite to flow of -ve charge





Current Density (J):



Area = A of Cross section

$$\vec{J} = \frac{i}{A}\hat{n}$$

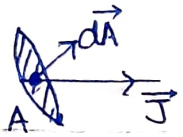
$\hat{n}$ → Unit Vector along i

→ Unit = Amp/m^2
→ Dimension = $[IL^{-2}]$
→ Vector Quantity
→ Direction: Along Current

If $\vec{J}$ = uniform

$$i = JA\cos\theta = \vec{J}\cdot\vec{A}$$

If $\vec{J}$ ≠ Uniform we take small elemental areas dA & find the current di through this small dA element and then integrate di



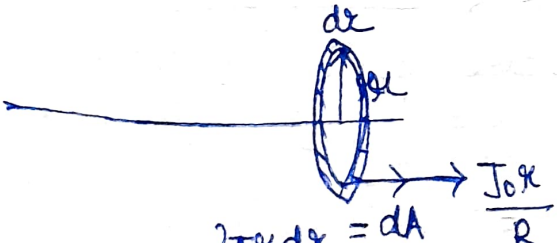
$$i = \int di = \int \vec{J}\cdot d\vec{A}$$

$$\therefore \boxed{i = \int \vec{J}\cdot d\vec{A}}$$

★ Surface integral of current density is current

Q

① Find current passing through this cylindrical wire.



$$di = \frac{J_0 r}{R}(2\pi r\,dr)\cos 0^\circ$$

$$i = \int di = \frac{J_0 2\pi}{R}\int_0^R r^2\,dr$$

$$i = \frac{2\pi J_0}{R}\cdot\frac{R^3}{3}$$

$$\boxed{i = \frac{2\pi J_0 R^2}{3}}$$

② Also find Avg current density.

$$J_{Avg} = \frac{i}{A}$$

$$\frac{2\pi J_0 R^2}{3} = J_{Avg}\,\pi R^2$$

$$J_{avg} = \frac{2J_0}{3}$$

Conductors
↳ large no. of free e^-
↳ Charge Carriers
↳ Not bounded to nucleus
↳ Free to move in lattice of metal

★ Free $e^\ominus$ density (n) $= \frac{\text{No. of free } e^\ominus}{\text{Volume}}$

Q A metal has atomic mass no. 100 and each atom has two free $e^\ominus$. Density of metal is 10,000 kg/m³, then find free $e^\ominus$ density.

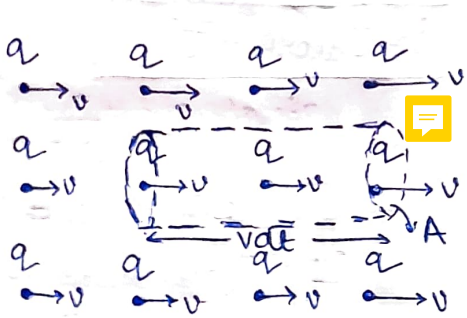
$$n = \frac{Ne}{\text{Volume}} = \frac{2N_{atom}}{\text{Volume}} = \frac{2(\text{no. of moles})\times N_A}{V} = \frac{2mN_A}{MV} = \frac{2\rho N_A}{M}$$

$$n = \frac{2\times 10^4 \times 6.023\times 10^{23}}{0.1} = 12.046\times 10^{28}\ m^{-3}$$

($M = 100$ g/mol, $M = 0.1$ kg/mol)

★ $n \propto$ Metallic nature

Relation between i and velocity of charge carriers:



$$i = \frac{nAv\,dt\,q}{dt}$$ {charge/time}

★ $i = nAvq$

★ $J = \frac{i}{A} = nvq$ $\quad \vec{J} = nq\vec{v}$ (use q with sign)

$$n = \frac{\text{no. of free } e^\ominus}{\text{Volume}}$$

A → Area of Cross-section
v → Velocity of charge carriers

Motion of free $e^\ominus$s in a conductor:
→ Free $e^\ominus$ will move with random velocities
→ Free $e^\ominus$ will move in every possible direction without giving preference to any direction, when no external EF is applied

→ $\vec{v_1}$, $\vec{v_2}$, $\vec{v_3}$, $\vec{v_n}$

$\vec{v}_{avg} = 0$ Average velocity is zero.

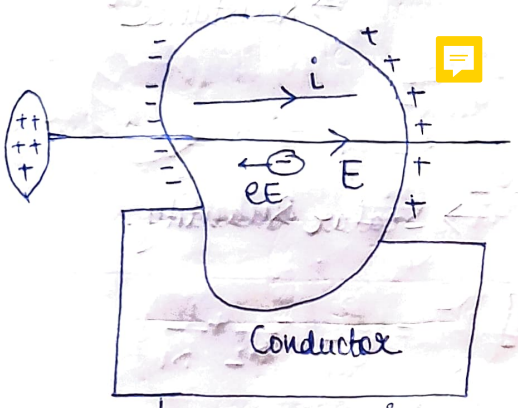
→ * $\frac{1}{2}mu^2 = \frac{3}{2}KT$ | K → Boltzman constant, T → Temperature (K) → At room temperature $u \approx 10^6$ m/s

→ Free $e^\ominus$ have velocity in order of 10^6 m/s but still through any cross-section charge passing will be zero and hence $i = 0$.

Charge going out of cross-section = Charge coming in of cross-section

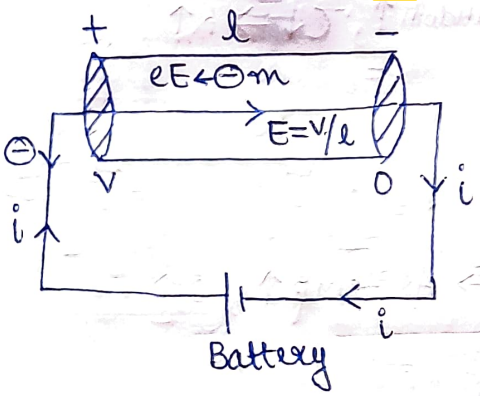
∴ Net $q = 0 \Rightarrow i = 0$

Now, we want the $e^\ominus$ to move in a certain direction,

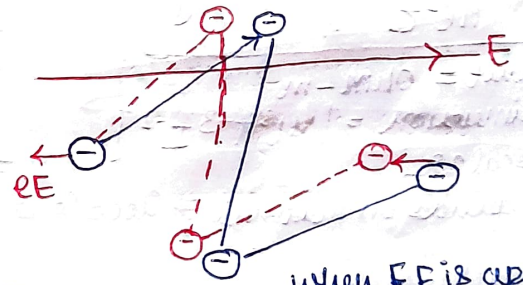


↳ Conducting wire

Here, we develop an external E.F with help of the charges. In presence of this field free $e^\ominus$ inside the conductor start moving opposite to the electric field. We can see that the $e^\ominus$ are now moving in a specific dirn. But as the charge is induced an EF inside the conductor also develops. When this EF becomes equal to Eext, Enet = 0 But we dont want this to happen. So we connect both ends to a conducting wire and form a closed loop so that charge does not stay at one place. This helps us developing current.



Inside the conductor
~~without~~ EF in blue
with EF in red



τ = Mean Relaxation time

↳ It is Avg of time interval b/w two collisions of an electron with lattice.

When EF is applied $e^\ominus$ start drifting in a particular direcn.

Now, for $e^\ominus$ $\vec{V} = \vec{u} + \vec{a}t$

So, $\vec{V_1} = \vec{U_1} + \frac{(-e)\vec{E}}{m} t_1$

$\vec{V_2} = \vec{U_2} + \frac{(-e)\vec{E}}{m} t_2$

⋮

$\vec{V_n} = \vec{U_n} + \frac{(-e)\vec{E}}{m} t_n$

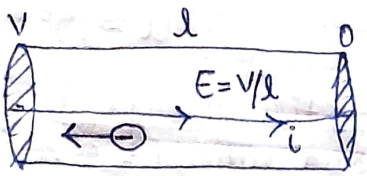
$\vec{V}_{avg} = \left\{\frac{\vec{U_1} + \vec{U_2} + \cdots + \vec{U_n}}{n}\right\} - \frac{e\vec{E}}{m}\left\{\frac{t_1 + t_2 + \cdots + t_n}{n}\right\}$

$\vec{U_1} + \vec{U_2} + \cdots + \vec{U_n} = 0$

* $\boxed{\vec{V}_{drift} = \vec{V}_{avg} = -\frac{e\vec{E}}{m}\tau}$

$\vec{V}_{drift}$ is opposite to $\vec{E}$.

Current in Conductor:



$$i = nAqV = nAe\frac{eE}{m}\tau$$

$$\boxed{i = \frac{nAe^2\tau}{m}E = \frac{nAe^2\tau}{m}\frac{V}{l}}$$

Now,

$$\boxed{V = \left(\frac{m}{ne^2\tau}\frac{l}{A}\right)i}$$

We know,

$$V = Ri$$

$$R = \frac{V}{i}$$

$$\therefore \boxed{R = \left(\frac{m}{ne^2\tau}\frac{l}{A}\right)}$$

↳ Resistance

Resistance:

$$R = \left(\frac{m}{ne^2\tau}\frac{l}{A}\right)$$

→ Unit = $\frac{\text{Volt}}{\text{Ampere}}$ = Ohm (Ω) → Scalar Quantity.

→ Dimension = $\frac{ML^2T^{-2}}{ITL} = [ML^2T^{-3}I^{-2}]$

→ Depends on shape size and geometry of wire.

→ Depends on mettalic nature. As mettalic nature ↑, n↑ ⇒ R↓

→ Depends on temperature. As T↑, v_{random}↑, τ↓ ⇒ R↑

Now, we can write $\boxed{R = \rho\frac{l}{A}}$

$$\boxed{\rho = \frac{m}{ne^2\tau} = \text{Resistivity}}$$

→ Unit = ohm-m → As temp ↑, ρ↑

→ Dimension = $[ML^3T^{-3}I^{-2}]$

→ Scalar

→ Depends on material ⇒ Good Conductor, n↑ ρ↓

Conductance (S):

$$i = \frac{nAe^2\tau}{m}\frac{V}{l} = \left(\frac{ne^2\tau}{m}\frac{A}{l}\right)V$$

$$\therefore \boxed{i = SV}$$

$$\therefore \boxed{S = \left(\frac{ne^2\tau}{m}\right)\frac{A}{l} = \frac{1}{R}}$$

→ Unit = $\frac{\text{Amp}}{\text{Volt}} = (\text{Ohm})^{-1}$ = mho = Siemens

→ Dimension = $[M^{-1}L^{-2}T^3I^2]$

Now can write $S = \sigma \frac{A}{l}$

$\sigma = \frac{ne^2\tau}{m}$ = conductivity

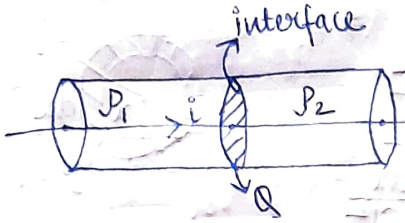
→ unit = $(\text{ohm-m})^{-1}$

→ Dimension = $[M^{-1}L^{-3}T^3I^2]$

Now, we know $i = \frac{nAe^2\tau E}{m} \Rightarrow \frac{i}{A} = \left(\frac{ne^2\tau}{m}\right)E$

$\therefore \vec{J} = \left(\frac{ne^2\tau}{m}\right)\vec{E}$ $\quad \vec{J} = \sigma\vec{E}$ $\quad \vec{J} = \frac{1}{\rho}\vec{E}$ $\quad \vec{E} = \vec{J}\rho$

**Q



=Find charge at interface

By definition: $\phi = \oint \vec{E} \cdot d\vec{s}$

$\phi = E_2A - E_1A$

$= (E_2 - E_1)A$

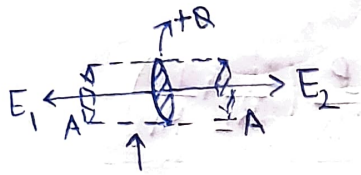
$= \left(\frac{i}{A}\rho_2 - \frac{i}{A}\rho_1\right)A$

$\phi = i(\rho_2 - \rho_1)$ — (1)

By Gauss law: $\phi = \frac{Q}{\varepsilon_0}$ — (2)

Now (1) = (2)

$Q = i\varepsilon_0(\rho_2 - \rho_1)$



We know $E = \rho J$

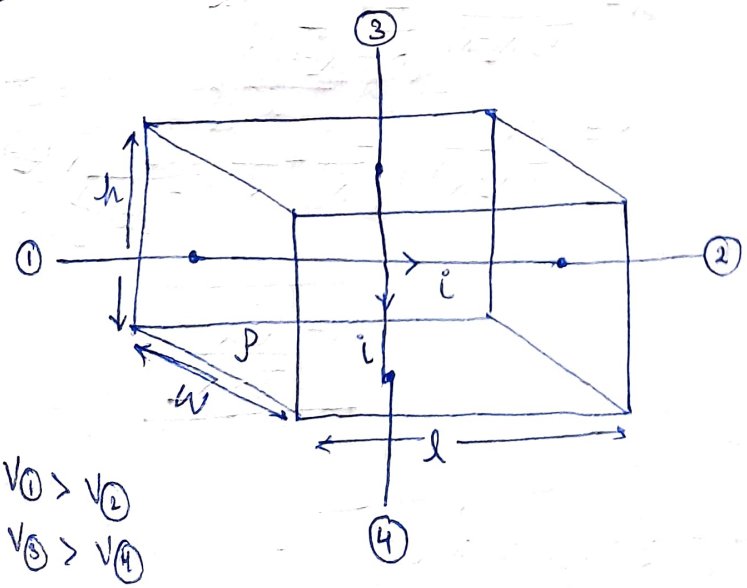
As ρ is different so E is also different in both parts.

Note:

$R = \frac{\rho l}{A}$ $\quad l \rightarrow$ length along current

$A \rightarrow$ Cross sectional area $\perp$ to current.

Ex



Find R_{12} and R_{34}

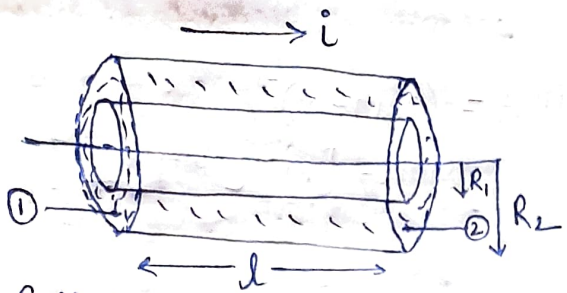
$R_{12} = \frac{\rho l}{wh}$

$R_{34} = \frac{\rho h}{wl}$

$V_1 > V_2$

$V_3 > V_4$

Q



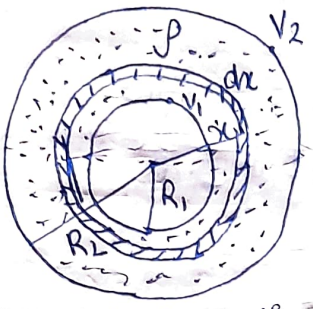
Cylindrical conductor with cavity (ρ)

Find R_{12}

$$R_{12} = \frac{\rho l}{\pi R_2^2 - \pi R_1^2}$$

$$\therefore \boxed{R_{12} = \frac{\rho l}{\pi (R_2^2 - R_1^2)}}$$

Q*

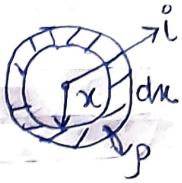


2 concentric conducting spheres

Also, $V_1 > V_2$ (Given)

Find resistance of the medium.

$$dR = \frac{\rho\, dx}{4\pi x^2}$$

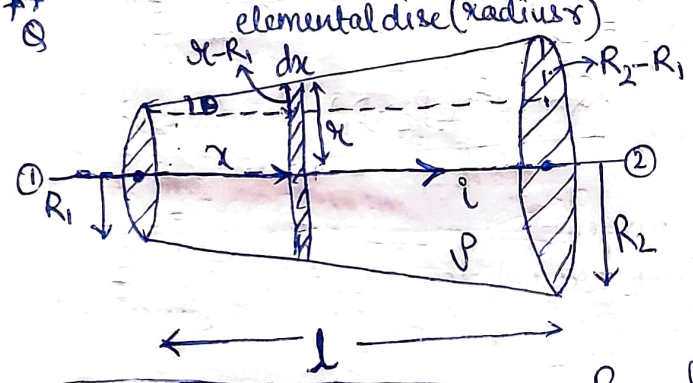


$$R = \int dR = \frac{\rho}{4\pi} \int_{R_1}^{R_2} \frac{dx}{x^2}$$

$$R = \frac{\rho}{4\pi} \left(\frac{-1}{x}\right)_{R_1}^{R_2} = \frac{\rho}{4\pi} \left(\frac{-1}{R_2} + \frac{1}{R_1}\right)$$

$$\boxed{R = \frac{\rho (R_2 - R_1)}{4\pi R_1 R_2}}$$

Q**



Find R_{12}

$$dR = \frac{\rho\, dx}{\pi r^2} = \frac{\rho\, dx}{\pi \left(R_1 + \frac{(R_2 - R_1)x}{l}\right)^2}$$

$$R = \int dR = \frac{\rho}{\pi} \int_0^l \frac{dx}{\left(R_1 + \frac{(R_2 - R_1)x}{l}\right)^2} \quad \text{—①}$$

$$\tan\theta = \frac{r - R_1}{x} = \frac{R_2 - R_1}{l}$$

$$\boxed{r = R_1 + \frac{(R_2 - R_1)x}{l}}$$

$$R = \frac{\rho}{\pi} \frac{l}{R_2 - R_1} \left(\frac{-1}{R_1 + \frac{(R_2 - R_1)x}{l}}\right)_0^l$$

$$R = \frac{\rho l}{\pi (R_2 - R_1)} \left(\frac{-1}{R_2} + \frac{1}{R_1}\right) = \frac{\rho l}{\pi (R_2 - R_1)} \times \frac{R_2 - R_1}{R_1 R_2}$$

$$\boxed{R = \frac{\rho l}{\pi R_1 R_2}}$$

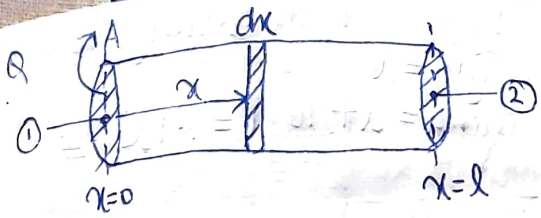
$$\int \frac{dx}{\left(R_1 + \frac{(R_2 - R_1)x}{l}\right)^2} = \frac{l}{R_2 - R_1} \int \frac{dt}{t^2}$$

Let $R_1 + \frac{(R_2 - R_1)x}{l} = t$

$\frac{R_2 - R_1}{l} dx = dt$

$$= \frac{l}{R_2 - R_1} \left(\frac{-1}{t}\right)$$

$$= \frac{l}{R_2 - R_1} \left(\frac{-1}{R_1 + \frac{(R_2 - R_1)x}{l}}\right) \{\text{Put in ①}\}$$



Given ρ varies as $\rho = \rho_0 e^{x/l}$

Find R_{12}

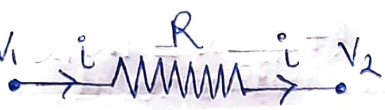
$$dR = \frac{\rho_0 e^{x/l}\, dx}{A}$$

$$R = \int dR = \frac{\rho_0}{A}\int_0^l e^{x/l}\, dx = \frac{\rho_0 l}{A}\left(e^{x/l}\right)_0^l$$

$$\boxed{R = \frac{\rho_0 l}{A}(e-1)}$$

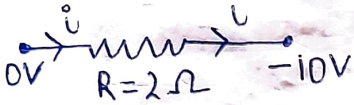
Ohm's law:

$(V_1 > V_2)$



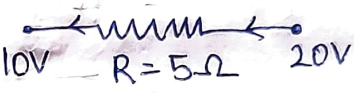
→ Current always flows from higher potential to lower potential in a resistor.

$$\boxed{i = \frac{\Delta V}{R} = \frac{V_1 - V_2}{R}}$$

Q 

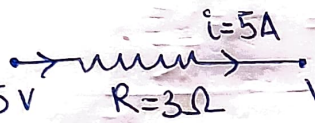
Find i

$$i = \frac{0-(-10)}{2} = 5A$$

Q 

Find i

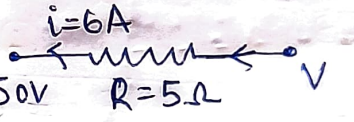
$$i = \frac{20-10}{5} = 2A$$

Q 

Find V.

$$i = \frac{5-V}{3} \Rightarrow 5\times 3 = 5-V$$

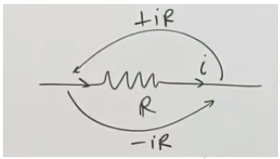
$$\boxed{V = -10 \text{ Volts}}$$

Q 

Find V

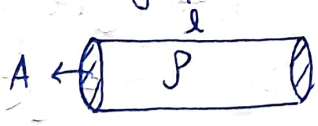
$$i = \frac{V-50}{5} \Rightarrow 6\times 5 = V-50$$

$$\boxed{V = 80 \text{ Volts}}$$



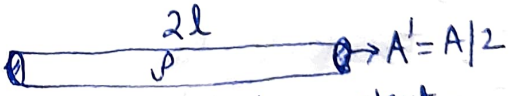
* Note: If we move in direction of current, potential decreases by iR.
If we move opposite to direction of current, potential increases by iR.

Dependency of Resistance on l and A:



$$R = \frac{\rho l}{A}$$

stretch →



Volume must remain constant.

$V = Al$ (old vol.) $\quad V = A'(2l)$

$$Al = A'(2l) \Rightarrow A' = A/2$$

$$R' = \frac{\rho(2l)}{A/2} = \frac{4\rho l}{A} = 4R \quad \therefore \boxed{R' = 4R}$$

So, in general, for stretching,

$$R = \frac{\rho l}{A} = \frac{\rho l}{(\text{Vol}/l)} = \frac{\rho l^2}{\text{Vol}}$$

$$\therefore \boxed{R \propto l^2} \text{ for stretching}$$

Q A wire is stretched to increase its length by 10%. Find % change in R.

$R = \frac{\rho l}{A} = \frac{\rho l^2}{\text{Volume}}$

Old length $= l$

New length $= l + \frac{10}{100} l = 1.1l$

$R' = \frac{\rho (1.1l)^2}{\text{Volume}} = 1.21 \frac{\rho l^2}{\text{Vol}} = 1.21R.$ $\therefore \boxed{R' = 1.21R}$

% Change in $R = \frac{R' - R}{R} \times 100 = \frac{1.21R - R}{R} \times 100 = 21\%$

Q If length of wire is stretched by 1%. Find % change in R.

Method 1:

$R = \frac{\rho l}{A} = \frac{\rho l^2}{\text{Vol.}}$ New length $= l + \frac{1}{100} l = 1.01l$

$R' = \frac{\rho (1.01l)^2}{\text{Vol}} = \frac{\rho}{\text{Vol.}} \left(1 + \frac{1}{100}\right)^2 l^2 = \frac{\rho}{\text{Vol}} \underbrace{\left(1 + \frac{2}{100}\right) l^2}_{\text{using binomial approx.}} = \left(1 + \frac{2}{100}\right) \frac{\rho l^2}{\text{Vol}} = \left(1 + \frac{2}{100}\right) R$

% change in $R = \frac{R' - R}{R} \times 100 = \frac{\left(1 + \frac{2}{100}\right) R - R}{R} \times 100 = 2\%$

Method 2:

only applicable for small changes.

$R = \frac{\rho l^2}{\text{Vol}}$

$\ln R = \ln \rho + 2 \ln l - \ln(\text{Vol})$

Differentiate

$\frac{dR}{R} = \frac{d\rho}{\rho}^{0} + 2\frac{dl}{l} - \frac{dV}{V}^{0} \Rightarrow \frac{dR}{R} = 2\left(\frac{dl}{l}\right)$

% change $\Rightarrow \left(\frac{dR}{R} \times 100\right) = 2\left(\frac{dl}{l} \times 100\right) = 2(1\%) = 2\%$

Dependency of Resistance on temperature:

As $T \uparrow$, $R \uparrow$.

$\alpha \rightarrow$ Thermal coeff of Resistance

Unit $\rightarrow$ $1/^\circ C$ or $1/\text{Kelvin}$

Dimension $= [K^{-1}]$

Let at $0^\circ C$, Resistance is R_0

Let at $T^\circ C$, Resistance is R

After some time, Temp $= (T + dT)^\circ C$, Resistance $= R + dR$

So, ① $dR \propto dT$

② $dR \propto R$

$\therefore \boxed{dR = \alpha R\, dT}$

$\int_{R_0}^{R} \frac{dR}{R} = \alpha \int_{0}^{\Delta T} dT$

$= \ln \frac{R}{R_0} = \alpha(\Delta T)$

$\therefore \boxed{R = R_0 e^{\alpha \Delta T}}$

We know $e^x = 1 + x + \dots$

If $x \to 0$, $e^x = 1 + x$

So, if $\alpha \Delta T \to$ small

★ $\boxed{R = R_0(1 + \alpha \Delta T)}$

★ $\boxed{\Delta R = R - R_0 = R_0 \alpha \Delta T}$

$R_0 \to$ Initial Resistance (Generally at 0°C)

$\alpha \to$ Coeff of thermal Resistance

$\Delta T \to$ Change in temperature.

Q At $0°C \Rightarrow R = 20\Omega$. Given $\alpha = 10^{-4}/°C$. Find Resistance at 20°C

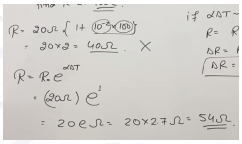
$\alpha \Delta T = 10^{-4} \times 20 = 2 \times 10^{-3}$ (small)

$R = R_0(1 + \alpha \Delta T) = 20(1 + 2 \times 10^{-3}) = 20 + 40 \times 10^{-3} = 20.04\Omega$

Q At $0°C \Rightarrow R = 20\Omega$. Given $\alpha = 10^{-2}/°C$. Find resistance at 100°C

$\alpha \Delta T = 10^{-2} \times 100 = 1$ (Not tending towards zero)

$R = R_0 e^{\alpha \Delta T} = 20e^1 = 20 \times 2.7 = 54\Omega$



Electrical Power: It is rate at which electrical energy is created or consumed.

$\boxed{\text{Power} = P = \frac{U}{t}}$

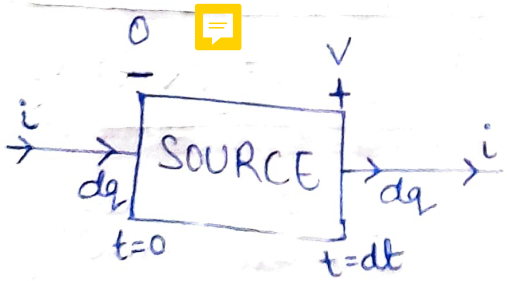
→ Unit = Joule/sec = Watt

→ Dimension = $[ML^2T^{-3}]$

Source: It converts some form of energy into electrical energy.

Ex Solar cells: Solar energy → Electrical energy

Chemical Cells: Chemical energy → Electrical energy



$U_i = dq \times 0 = 0$

$U_f = dq \times V = Vdq$

Electrical energy increased = dU

$dU = Vdq$

Power of Source = $\frac{dU}{dt} = \frac{dq}{dt} V = iV$

$\therefore \boxed{\text{Power} = Vi}$

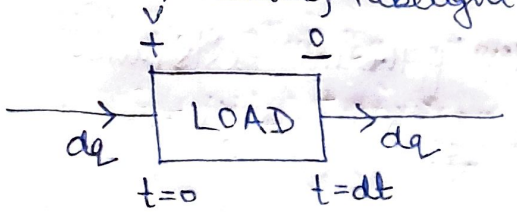
V → Potential difference across source

i → Current.

→ In source, current enters at lower potential and exits at higher potential.

Load: It consumes electrical energy and converts it into some other form of energy.

Ex – Fan, heater, tubelight, AC



$U_i = dq\,V = V\,dq$

$U_f = dq\,0 = 0$

Energy consumed by load $= dU = dq\,V$

Power of load $= \frac{dU}{dt} = \frac{dq}{dt} V = iV$

$\boxed{P = Vi}$

→ V = Potential difference across load

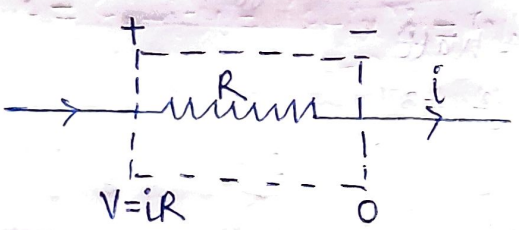
→ i = Current.

→ In load, current enters at higher potential and exists at lower potential.

Electrical Power in Resistor.

→ Resistor always acts like load. since current always enters through high potential and exits through low potential

→ It consumes electrical energy and converts into Heat / light.



$P = Vi = (iR)i = i^2R = \frac{V^2}{R}$

$\therefore \boxed{P = Vi = i^2R = \frac{V^2}{R}}$

Heat / Energy dissipated by resistor:

If P = constant

$\boxed{H = Pt = Vit = i^2Rt = \frac{V^2}{R}t}$

If P ≠ constant.

$\boxed{H = \int P\,dt = \int Vi\,dt = \int i^2R\,dt = \int \frac{V^2}{R}\,dt}$

Unit of electrical energy = Joule

1 unit = 1 KW hr

= 1000 W × 3600 s

$\boxed{1 \text{ unit} = 3.6 \times 10^6 \text{ J}}$

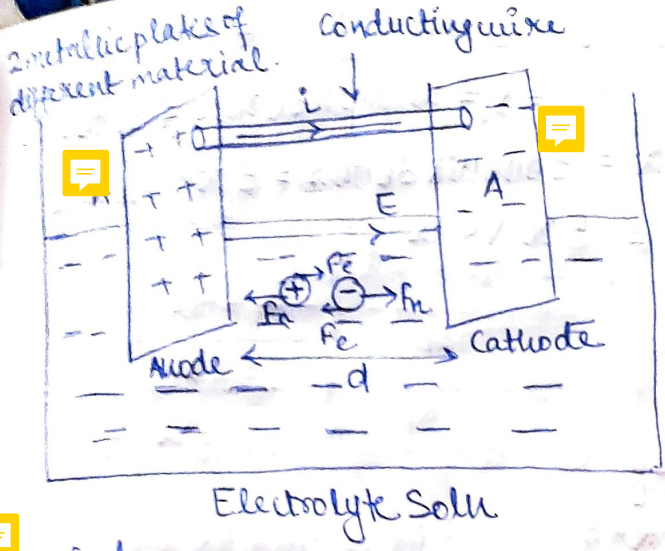
Ex If 50 units energy is consumed then energy = $50 \times 3.6 \times 10^6$ J

Battery:

→ Used to provide constant potential difference in a conducting wire.

→ Converts chemical energy into electrical energy {in general}

→ Mostly used as a source.



Initially the cations move towards anode and the anions move towards cathode. Slowly an $\vec{E}$ develops b/w the plates. When $\vec{F}$ gets equal to $\vec{F}_n$, no more charges move now. Now, we connect the plates with a conducting wire. A current flows from Anode to cathode. -ve charge flows through the wire from cathode to anode and neutralise some +ve charge. When this happens $|\vec{E}|\downarrow$, due to which $F_n > F_e$ and charges again begin to flow and the same condⁿ is achieved. This process continues till all the cations and anions in electrolyte is consumed. When this happens, we say battery is discharged.

$W = f_n d$

Electromotive Force $= \mathcal{E} = \frac{W}{q} = \frac{F_n d}{q}$

($F_n \rightarrow$ Unknown force)

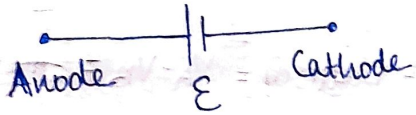
→ Resistance of battery itself is known as internal resistance (r).

① $r \propto d$ (sep b/w plates) ② $r \propto \frac{1}{\text{Area of plates}}$ ③ $r \propto$ Concentration of electrolyte soln

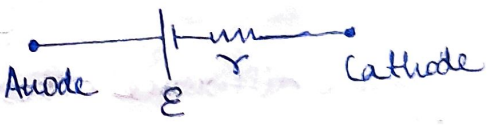
④ $r \propto \frac{1}{\text{temp. of electrolyte soln}}$

Representation of Battery:

Ideal Battery
→ Has no internal resistance



Non-ideal Battery
→ Has internal resistance



Electromotive Force (EMF):

→ Not a force

→ Emf is capability of a battery to flow charge in a circuit.

→ Emf is work done by battery on charge carriers per unit charge.

$$\mathcal{E} = \frac{W_{battery}}{Q}$$

→ Unit = J/Coulomb = Volt

→ Dimension $= \frac{ML^2T^{-2}}{IT} = [ML^2T^{-3}I^{-1}]$

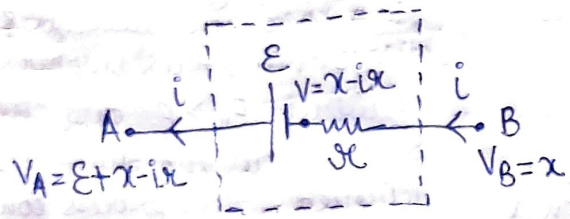
Given, Battery:- Rated as 10 Amp-hr and 5kW hr

⇒ Total Charge that battery can provide = $Q = 10\,Amp \times 3600\,s = 36000\,C$

Total Work Done by battery = $W = 5000W \times 3600\,s = 5000 \times 3600\,J$

$$\mathcal{E} = \frac{W}{Q} = \frac{5000 \times 3600}{36000}\,\frac{J}{C} = 500\ Volts$$

Battery as Source:



Potential difference across terminals of battery = $V_A - V_B = \mathcal{E} - ir$

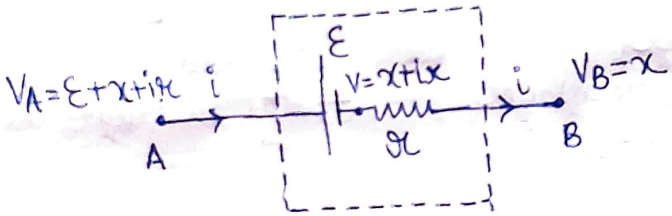
Rate at which chemical energy is consumed = $\frac{dW}{dt} = \frac{\mathcal{E}\,dq}{dt} = \mathcal{E}i$

Rate at which energy is lost in battery itself = i^2r

$$P_{out} = \mathcal{E}i - i^2r = (\mathcal{E} - ir)i = (V_A - V_B)i$$

$$\therefore \boxed{P_{out} = (V_A - V_B)i}$$

Battery as a load:



Potential difference across terminals of battery = $V_A - V_B = \mathcal{E} + ir$

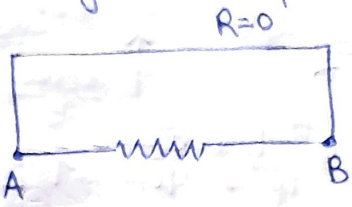
Rate at which chemical energy is stored = $\frac{dW}{dt} = \frac{\mathcal{E}\,dq}{dt} = \mathcal{E}i$

Rate at which power is lost by internal resistance = i^2r

Power input = $P_{in} = \mathcal{E}i + i^2r = (\mathcal{E} + ir)i = (V_A - V_B)i$

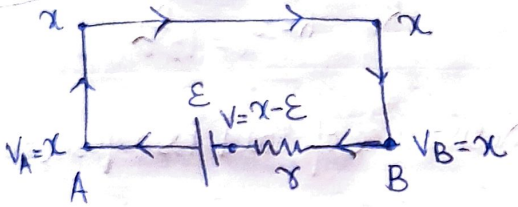
$$\boxed{P_{in} = (V_A - V_B)i}$$

Short Circuiting: Short circuiting two points A and B, means that connecting these points A and B by a wire of zero resistance.



$$V_A - V_B = 0$$

Short-Circuiting a battery:

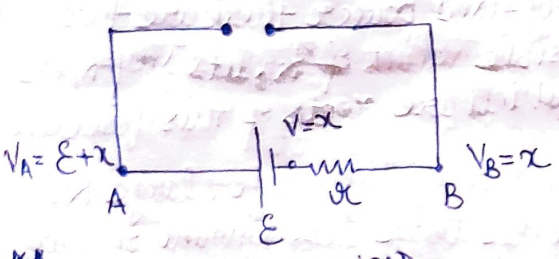


applying Ohm's law across the internal resistance r :

$$i = \frac{x-(x-\varepsilon)}{r} = \frac{\varepsilon}{x} \quad \therefore \boxed{i = \frac{\varepsilon}{x}}$$

$$\boxed{V_{AB} = V_A - V_B = 0}$$

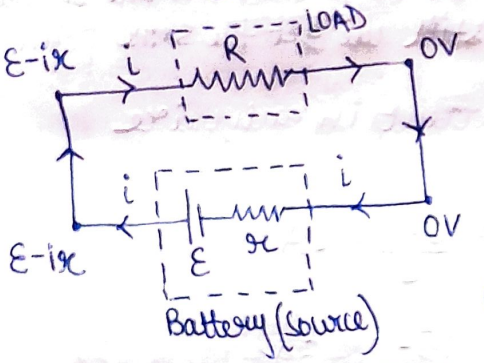
Open-circuit:



$$\boxed{V_A - V_B = \varepsilon}$$

★ If there is no current through battery, the ε of battery is equal to potential difference across the terminals of battery.

★★★ **Ex**



① Current in Circuit $= i = \frac{\Delta V}{R} = \frac{\varepsilon - ix}{R}$

$$\therefore \boxed{i = \frac{\varepsilon}{R+x}}$$

② Potential difference across terminals of battery $= \varepsilon - ix = \varepsilon - \frac{\varepsilon}{R+x}x = \frac{\varepsilon R}{R+x}$

③ Rate at which chemical energy is consumed $= \varepsilon i = \frac{\varepsilon^2}{R+x}$

④ Rate at which energy is lost by battery itself $= i^2 x = \frac{\varepsilon^2 x}{(R+x)^2}$

⑤ P_{out} = Power consumed by load $= \frac{\varepsilon^2}{R+x} - \frac{\varepsilon^2 x}{(R+x)^2} = \frac{\varepsilon^2 R}{(R+x)^2} = i^2 R$

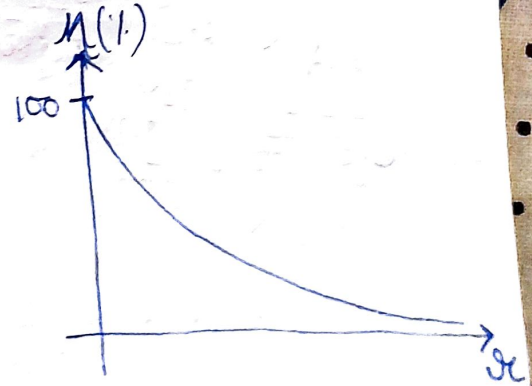
⑥ Efficiency $(\eta) = \eta = \frac{P_{out}}{P_{in}} \times 100$

$$\eta = \frac{P_{out}}{\text{Rate at which C.E is consumed}} \times 100 = \frac{i^2 R}{\varepsilon i} \times 100$$

$$\boxed{\eta = \frac{R}{R+x} \times 100}$$

$x = 0 \Rightarrow \eta = 100\% \Rightarrow$ ideal battery

$x \uparrow, \eta \downarrow$



⑦ Graph b/w P and R:

$$P = \frac{\mathcal{E}^2 R}{(R+r)^2}$$

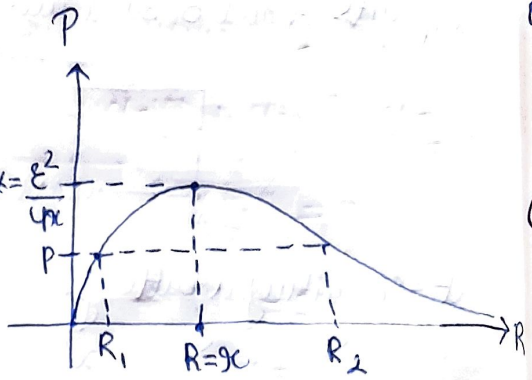
If $R=0 \Rightarrow P=0$

If $R \to \infty \Rightarrow \frac{\mathcal{E}^2 R}{R^2} = \frac{\mathcal{E}^2}{R} \to 0$

$$\frac{dP}{dR} = \mathcal{E}^2 \left\{ \frac{(R+r)^2(1) - R(2)(R+r)}{(R+r)^4} \right\} = 0$$

$\Rightarrow R+r = 2R$

$\boxed{R=r}$ $\therefore \boxed{P_{max} = \frac{\mathcal{E}^2}{4r}}$



* Load will consume maxm power when Resistance of load is equal to internal resistance.

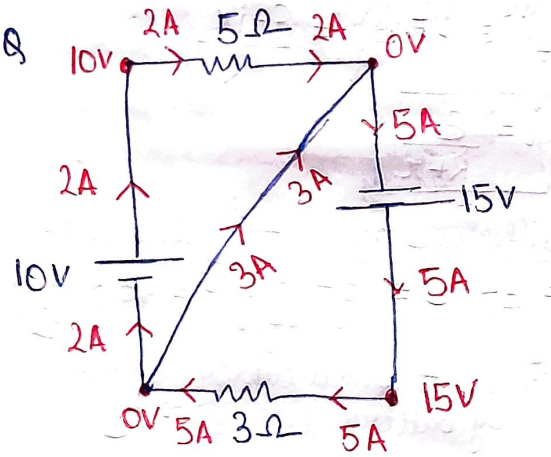
Now, $P = \frac{\mathcal{E}^2 R}{(R+r)^2} = \frac{\mathcal{E}^2 R}{R^2+r^2+2Rr} \Rightarrow PR^2 + R\{2Pr - \mathcal{E}^2\} + r^2 P = 0$

* For any fixed power there are two possible values of R, lets say R_1, R_2 which are roots of this quadratic.

* $\boxed{R_1 + R_2 = \frac{\mathcal{E}^2 - 2Pr}{P}}$ * $\boxed{R_1 R_2 = r^2}$

Out of these two values of R, the larger one is beneficial as for large R, current will be less, so battery will discharge less amount of charges & and can work for more time but still we get the same power.

Q

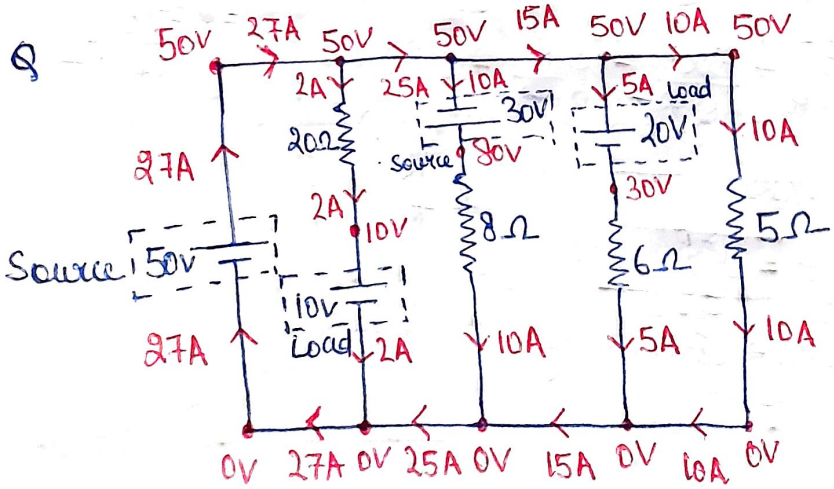


Find current in each wire

$i = \frac{10-0}{5} = 2A$

$i = \frac{15-0}{5} = 3A$

Q



Find current in each wire

Q Bulb is rated as 220V, 100 Watt. Find resistance of bulb.

220V, 100W means at 220V, bulb will give power as 100W.

$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{(220)^2}{100} = 484\,\Omega$$

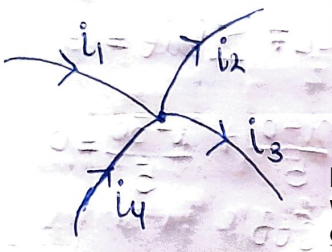
② Find power of this bulb if 110V of potential difference is applied across bulb.

$$P = \frac{V^2}{R} = \frac{(110)^2}{484} = 25\,W$$

Kirchoff's Current Law (KCL):

→ Also known as junction law

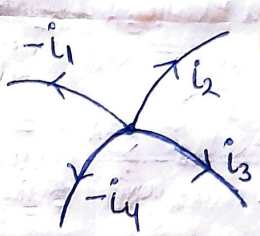
→ Based on charge conservation.



$$\sum i_{in} = \sum i_{out}$$

$$i_1 + i_4 = i_2 + i_3$$

however, for unknown currents, we won't know the direction of flow of current

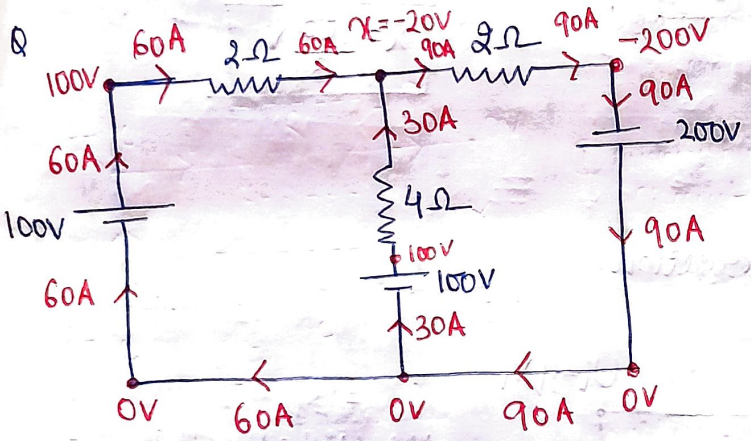


$$i_2 + i_3 + (-i_1) + (-i_4) = 0$$

$$\therefore \sum i_{out} = 0$$

hence, we adopt this method

Q



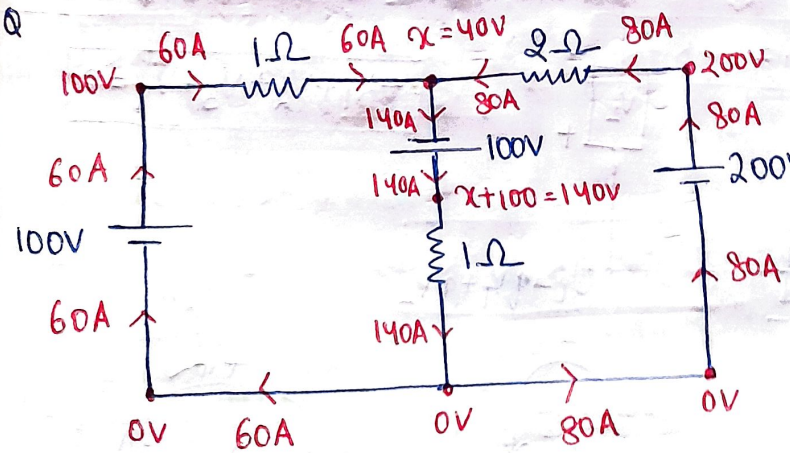
Find current in each wire using KCL to find x:

$$\frac{x-100}{2} + \frac{x-100}{4} + \frac{x+200}{2} = 0$$

$$5x = -100$$

$$\boxed{x = -20V}$$

Q



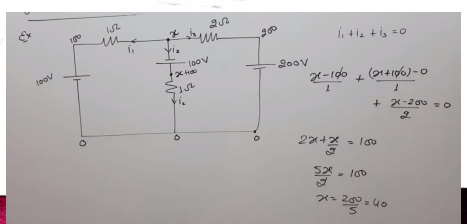
Find current in each wire

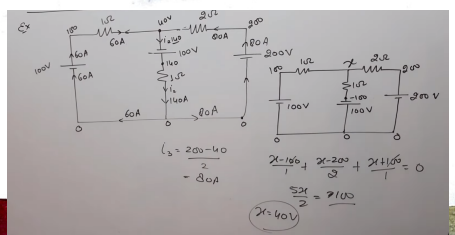
using KCL: $\sum i_{out} = 0$

$$\frac{x-100}{1} + \frac{x+100}{1} + \frac{x-200}{2} = 0$$

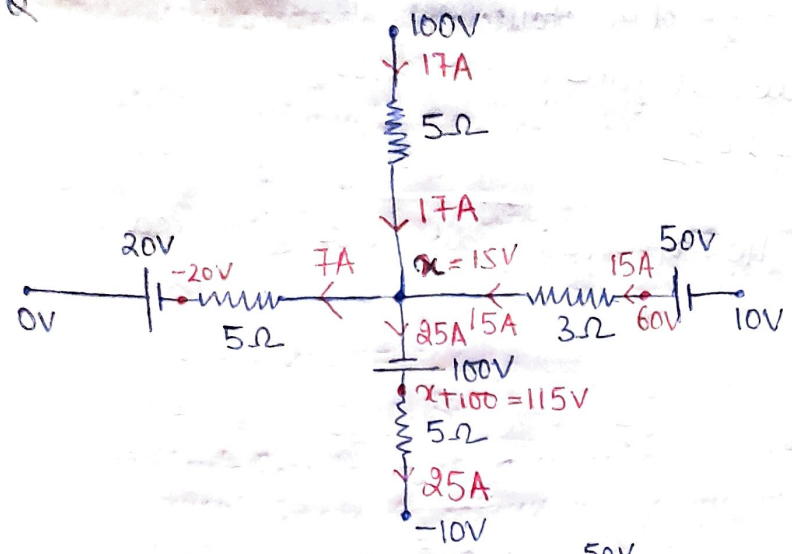
$$5x = 200$$

$$\boxed{x = 40V}$$





Q



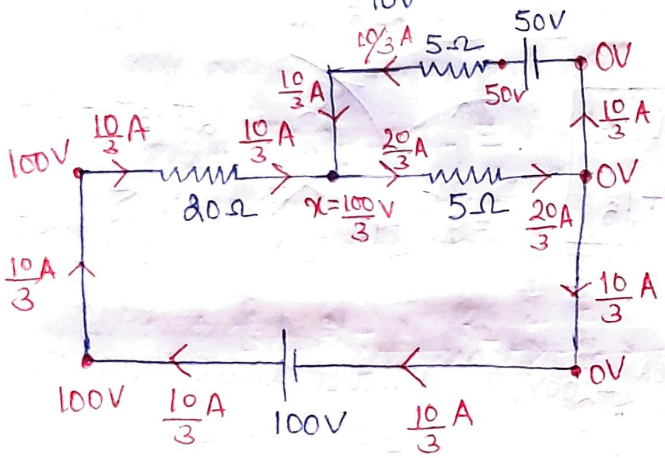
Find current in each wire using KCL: $\sum i_{out} = 0$

$$\frac{x-100}{5} + \frac{x-60}{3} + \frac{x+110}{5} + \frac{x+20}{5} = 0$$

$$\Rightarrow 14x = 210$$

$$\boxed{x = 15V}$$

Q



Find current in each wire using KCL: $\sum i_{out} = 0$

$$\frac{x-100}{20} + \frac{x-0}{5} + \frac{x-50}{5} = 0$$

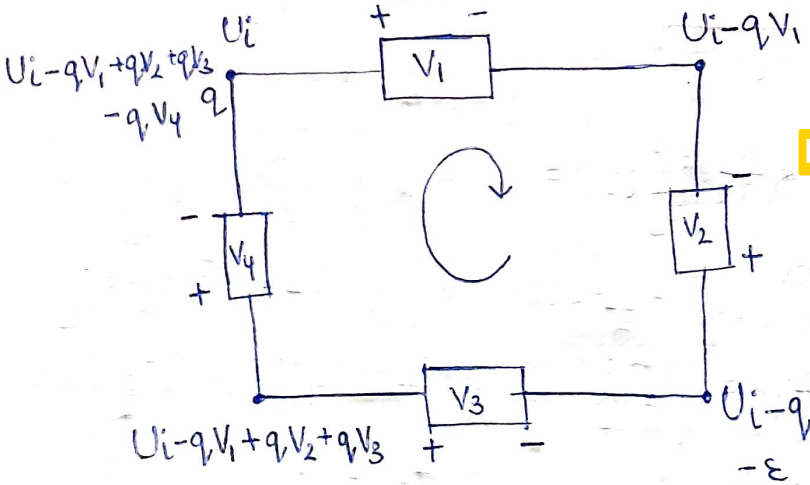
$$9x = 300$$

$$\boxed{x = \frac{100}{3} V}$$

Kirchoff's Voltage Law (KVL):

→ Loop law

→ It is based on energy conservation.

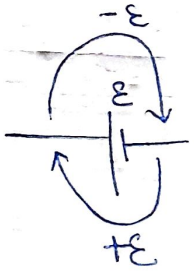


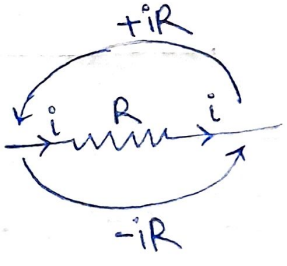
$$V_i = V_i - qV_1 + qV_2 + qV_3 - qV_4$$

$$\boxed{-V_1 + V_2 + V_3 - V_4 = 0}$$

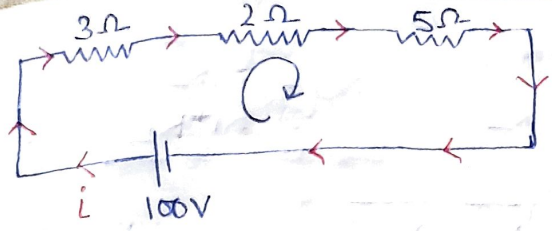
★ In an electrical circuit, algebraic sum of potential differences in a closed loop is always zero.

$$\boxed{\sum_{i=1}^{n} V_i = 0}$$





A



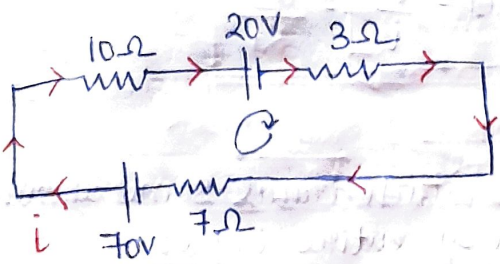
Find current in the circuit using KVL:

$100 - 3i - 2i - 5i = 0$

$100 = 10i$

$\boxed{i = 10A}$

Q



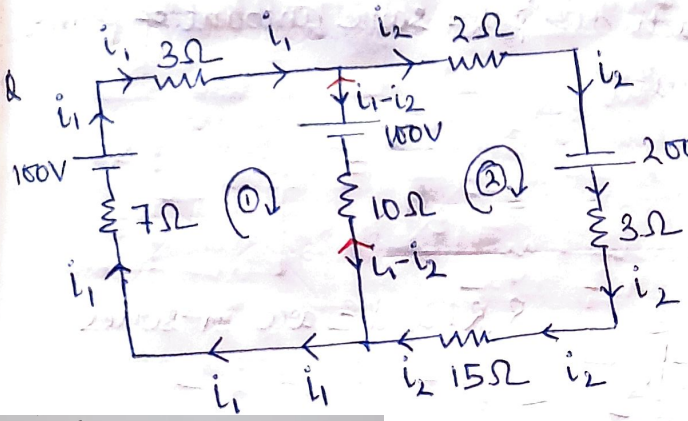
Find current in the circuit. using KVL:

$70 - 10i - 20 - 3i - 7i = 0$

$50 = 20i$

$\boxed{i = 2.5A}$

Q



Find current in the circuit. using KVL:

In loop 1:

$100 - 3i_1 - 100 - 10(i_1 - i_2) - 7i_1 = 0$

$-20i_1 + 10i_2 = 0$

$\boxed{i_2 = 2i_1}$ —①

In loop 2:

$200 - 3i_2 - 15i_2 + 10(i_1 - i_2) + 100 - 2i_2 = 0$

$10i_1 - 30i_2 = -300$

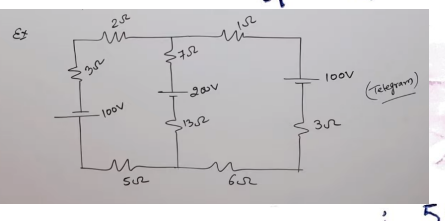
$10i_1 - 30(2i_1) = -300$

$-50i_1 = -300$

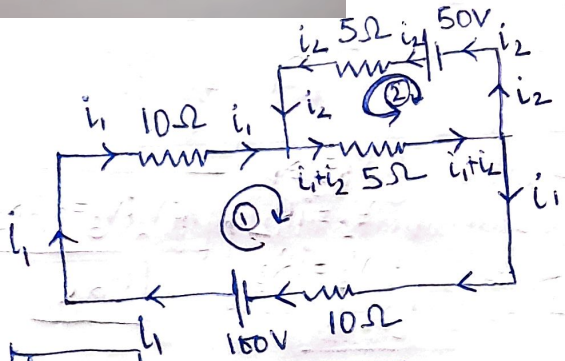
$\boxed{i_1 = 6A}$

$\boxed{i_2 = 12A}$

Ex



Q



Find current in each wire using KVL:

In loop ①:

$100 - 10i_1 - 5(i_1 + i_2) - 10i_1 = 0$

$5i_1 + i_2 = 20$ —①

In loop ②:

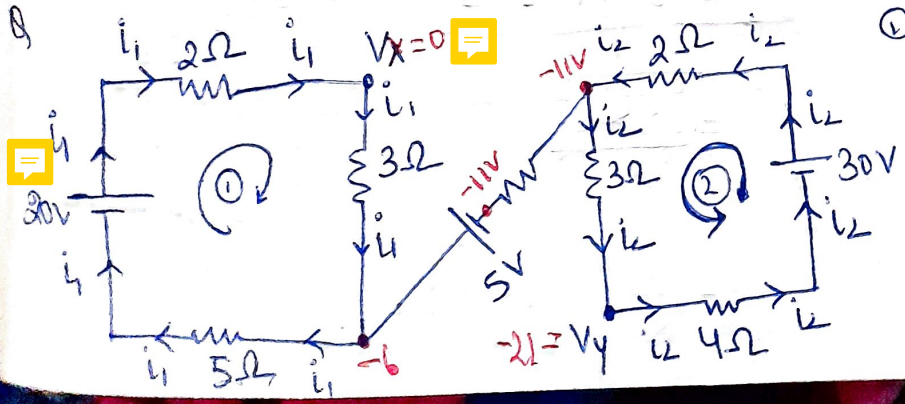
$50 - 5i_2 - 5(i_1 + i_2) = 0$

$i_1 + 2i_2 = 10$ —②

$\boxed{i_1 = \frac{10}{3}A}$

$\boxed{i_2 = \frac{10}{3}A}$

Q



① Find current in each wire using KVL:

In loop ①:

$20 - 2i - 3i_1 - 5i_1 = 0$

$\boxed{i_1 = 2A}$

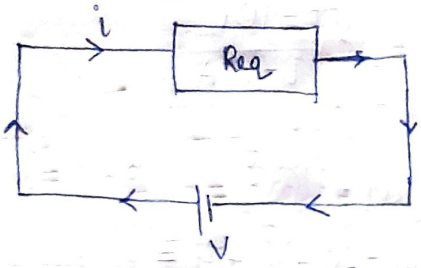
In loop ②

$30 - 2i_2 - 3i_2 - 4i_2 = 0$

$\boxed{i_2 = \frac{10}{3}A}$

② Find $V_x - V_y$: $V_x - V_y = 0 - (-21) = 21V$

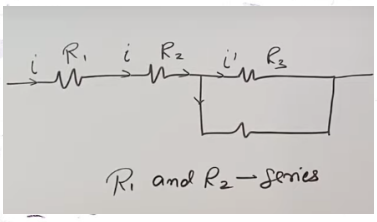
Equivalent Resistance:



$$R_{eq} = \frac{V}{i}$$

→ Equivalent Resistance is a resistance which draws same amount of current from a battery as drawn by entire circuit.

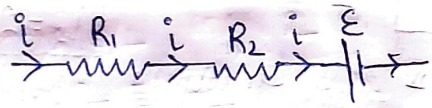
→ Equivalent is a resistance which dissipates same amount of power as dissipated by entire circuit.



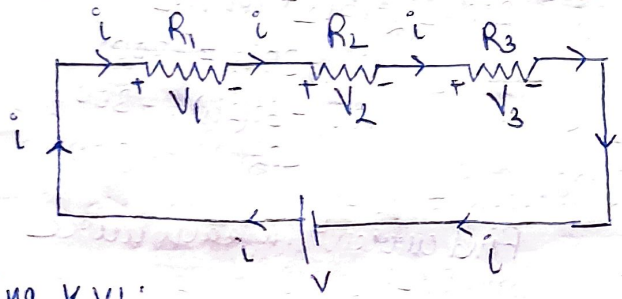
Series-Combination:

→ They must be joined end to end

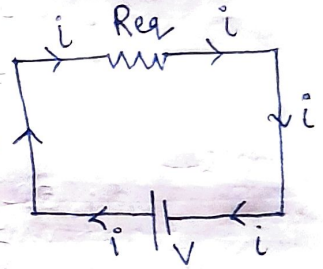
→ They must have same current.



R_1, R_2 and ε are in series



≡



Using KVL:

$V - V_1 - V_2 - V_3 = 0$

$V - iR_1 - iR_2 - iR_3 = 0$

$\frac{V}{i} = R_1 + R_2 + R_3 \rightarrow$ * $\boxed{R_{eq} = R_1 + R_2 + R_3}$

$V_1 = iR_1$

$V_2 = iR_2$

$V_3 = iR_3$

$\boxed{V_1 : V_2 : V_3 = R_1 : R_2 : R_3}$

$\boxed{V \propto R}$

Power dissipated:

$P_1 = i^2 R_1$

$P_2 = i^2 R_2$

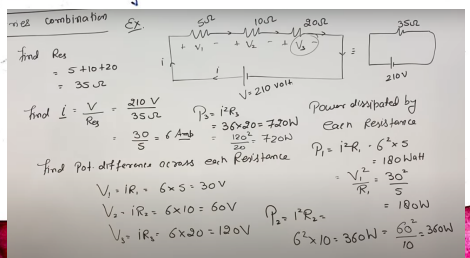
+ $P_3 = i^2 R_3$

$P_{net} = i^2 (R_1 + R_2 + R_3)$

$\therefore \boxed{P_{net} = i^2 R_{eq}}$

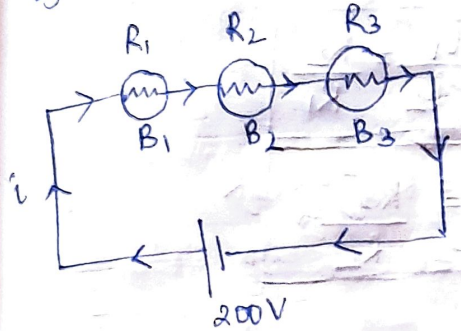
$\boxed{P_1 : P_2 : P_3 = R_1 : R_2 : R_3}$

$\boxed{P \propto R}$



Given 3 bulbs rated as
B_1: 200V, 25W
B_2: 200V, 50W
B_3: 200V, 100W

Find ① Resistance of each bulb
② Req of circuit.
③ Current in circuit
④ Potential difference across each bulb
⑤ Power of each bulb.



① $R_1 \Rightarrow P_1 = \frac{V^2}{R_1} \Rightarrow R_1 = \frac{V^2}{P_1} = \frac{200 \times 200}{25} = 1600\,\Omega$

$R_2 = \frac{V^2}{P_2} = \frac{200 \times 200}{50} = 800\,\Omega$

$R_3 = \frac{V^2}{P_3} = \frac{200 \times 200}{100} = 400\,\Omega$

② $R_{eq} = 1600 + 800 + 400 = 2800\,\Omega$

③ $i = \frac{V}{R_{eq}} = \frac{200}{2800} = \frac{1}{14}\,A$

④ $V_1 = iR_1 = \frac{1}{14} \times 1600 = \frac{800}{7}$

$V_2 = iR_2 = \frac{800}{14} = \frac{400}{7}$

$V_3 = iR_3 = \frac{400}{14} = \frac{200}{7}$

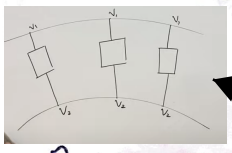
⑤ $P_1 = i^2 R_1 = \left(\frac{1}{14}\right)^2 \times 1600 = \frac{1600}{196}\,W$

$P_2 = i^2 R_2 = \left(\frac{1}{14}\right)^2 \times 800 = \frac{800}{196}\,W$

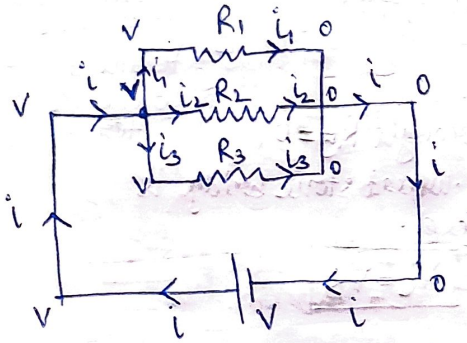
$P_3 = i^2 R_3 = \left(\frac{1}{14}\right)^2 \times 400 = \frac{400}{196}\,W$

$P_{net} = \frac{100}{7}\,W$

Also as $P \propto R$, B_1 will grow the brightest.



Parallel Combination:
→ Elements must have same potential difference across them.



Using KCL:
$i = i_1 + i_2 + i_3$

$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$

$\Rightarrow \boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$

$\boxed{S_{eq} = S_1 + S_2 + S_3}$ (Conductance)

$i = \frac{V}{R_{eq}}$

$i_1 = \frac{V}{R_1}$

$i_2 = \frac{V}{R_2}$

$i_3 = \frac{V}{R_3}$

$\boxed{i_1 : i_2 : i_3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}}$

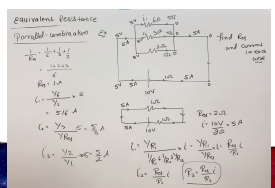
$\boxed{i_1 : i_2 : i_3 = S_1 : S_2 : S_3}$

$\boxed{i_1 + i_2 + i_3 = i}$

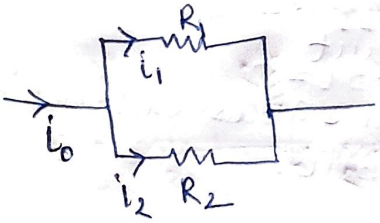
$\boxed{i_1 = \frac{1/R_1}{1/R_1 + 1/R_2 + 1/R_3}\, i = \frac{S_1}{S_1 + S_2 + S_3}\, i}$ $\quad \boxed{i_1 = \frac{R_{eq}\, i}{R_1}}$

$\boxed{i_2 = \frac{1/R_2}{1/R_1 + 1/R_2 + 1/R_3}\, i = \frac{S_2}{S_1 + S_2 + S_3}\, i}$ $\quad \boxed{i_2 = \frac{R_{eq}\, i}{R_2}}$

$\boxed{i_3 = \frac{1/R_3}{1/R_1 + 1/R_2 + 1/R_3}\, i = \frac{S_3}{S_1 + S_2 + S_3}\, i}$ $\quad \boxed{i_3 = \frac{R_{eq}\, i}{R_3}}$



Special Case: Two resistance connected in parallel.



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \boxed{R_{eq} = \frac{R_1 R_2}{R_1 + R_2}} \star$$

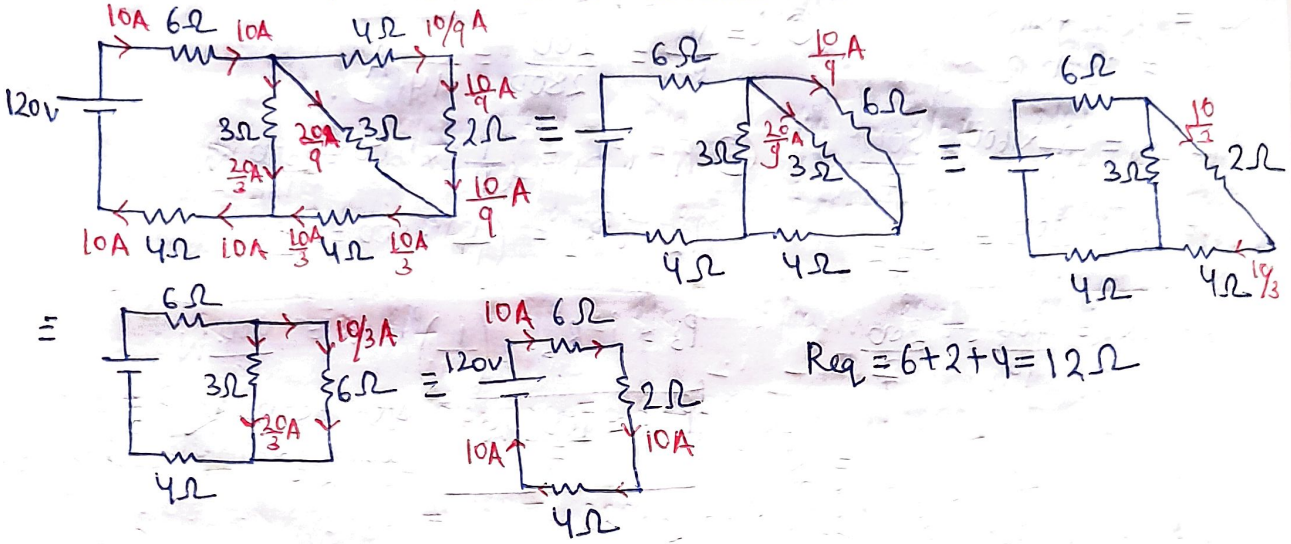
$i_1 + i_2 = i_0$ — (1)

$i_1 R_1 = i_2 R_2$ — (2) $\Rightarrow \boxed{\frac{i_1}{i_2} = \frac{R_2}{R_1}} \star$

$$\therefore \boxed{i_1 = \frac{R_2 i_0}{R_1 + R_2} = \frac{R_{eq} i_0}{R_1}}$$

$$\boxed{i_2 = \frac{R_1 i_0}{R_1 + R_2} = \frac{R_{eq} i_0}{R_2}}$$

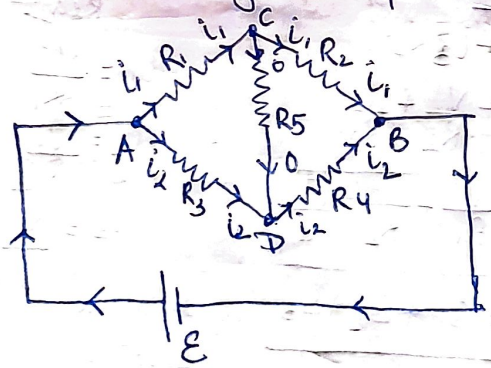
Q Find R_{eq}. Also find current in each wire.



$R_{eq} = 6 + 2 + 4 = 12\,\Omega$

Wheatstone Bridge:

→ It is an arrangement of 5 resistances and 4 terminals
- 2 terminals are connected to 2 terminals
- 2 terminals are connected to 3 terminals



★ If i through $R_5 = 0$, we say wheat stone bridge is balanced.

$\Rightarrow \boxed{V_C = V_D}$

$V_A - V_C = i_1 R_1$ — (1)

$V_C - V_B = i_1 R_2$ — (2)

$V_A - V_D = i_2 R_3$ — (3)

$V_D - V_B = i_2 R_4$ — (4)

$$\frac{V_A - V_C}{V_C - V_B} = \frac{R_1}{R_2} \text{ — (A)} \qquad \frac{V_A - V_D}{V_D - V_B} = \frac{R_3}{R_4} \text{ — (B)}$$

As $V_C = V_D \Rightarrow$ (A) = (B)

$$\therefore \star \boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}} \qquad \boxed{\frac{R_1}{R_3} = \frac{R_2}{R_4}}$$

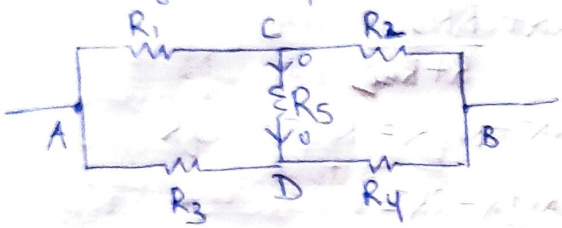
Condition for balanced bridge

$$\star \boxed{R_1 R_4 = R_2 R_3}$$

Now, as no current goes through R_5, we can remove it.

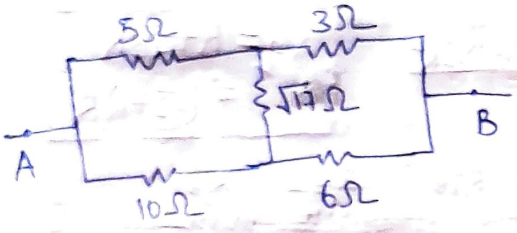
$$R_{eq} = \frac{(R_1+R_2)(R_3+R_4)}{R_1+R_2+R_3+R_4}$$

Alternate arrangement of wheatstone bridge:



$$R_1R_4 = R_2R_3$$

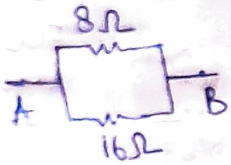
Q



Find $(R_{eq})_{AB}$.

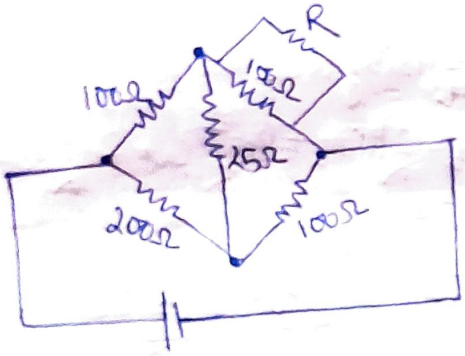
$R_1R_4 = R_3R_2 \Rightarrow 5\times6 = 3\times10$

$\therefore$ It is balanced wheatstone bridge.



$$R_{eq} = \frac{8\times16}{24} = \frac{16}{3}\ \Omega$$

Q



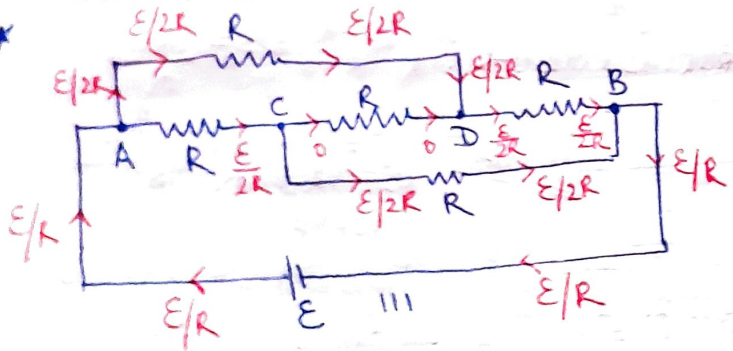
Find R such that current through 25Ω is zero.

$$100\times100 = 200\times\frac{100R}{100+R}$$

$$(100+R)\,50 = 100R \Rightarrow \frac{5000}{50} = R$$

$$R = 100\ \Omega$$

Q ***

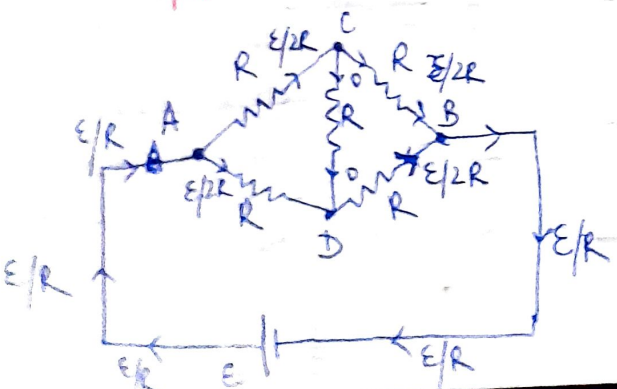


Find current in each wire.

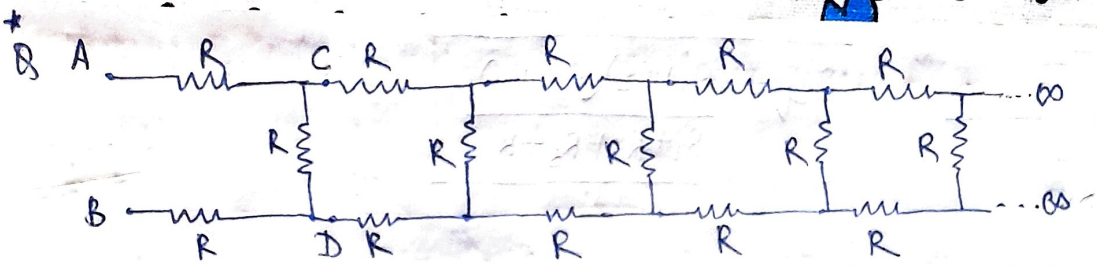
$R_1R_4 = R_3R_4$

$\therefore R^2 = R^2$

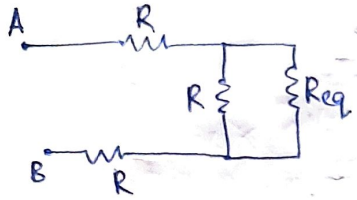
$$R_{eq} = \frac{4R^2}{4R} = R$$



Q



$R_{AB} = R_{eq} = R_{CD}$



$$R_{AB} = R_{eq} = 2R + \frac{R R_{eq}}{R + R_{eq}}$$

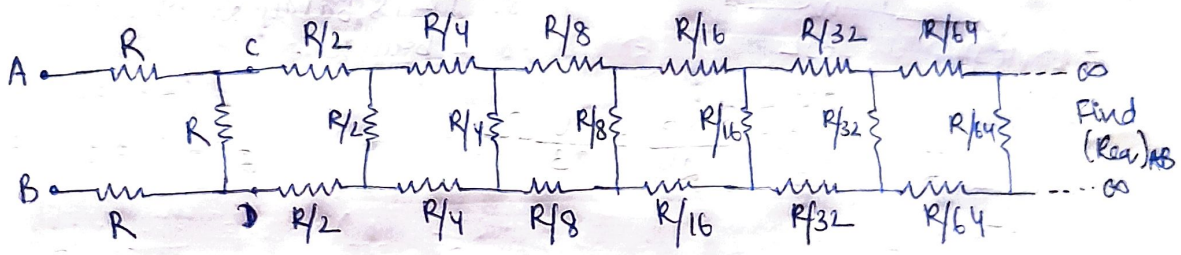
$$= 2RR_{eq} + 2R^2 + R_{eq}R = R_{eq}^2 + R_{eq}R$$

$$= R_{eq}^2 - 2R_{eq}R - 2R^2 = 0$$

$$R_{eq} = \frac{2R \pm \sqrt{4R^2 + 4 \times 2R^2}}{2}$$

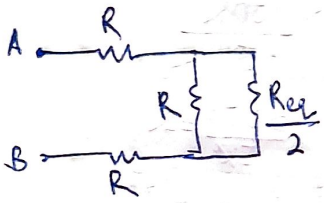
$$R_{eq} = R + \sqrt{3}R = (\sqrt{3}+1)R$$

Q



$R_{AB} = R_{eq}$

$R_{CD} = R_{eq}/2$



$$R_{AB} = R_{eq} = 2R + \frac{R R_{eq}}{2R + R_{eq}}$$

$$= 4R^2 + 2RR_{eq} + RR_{eq} = 2RR_{eq} + R_{eq}^2$$

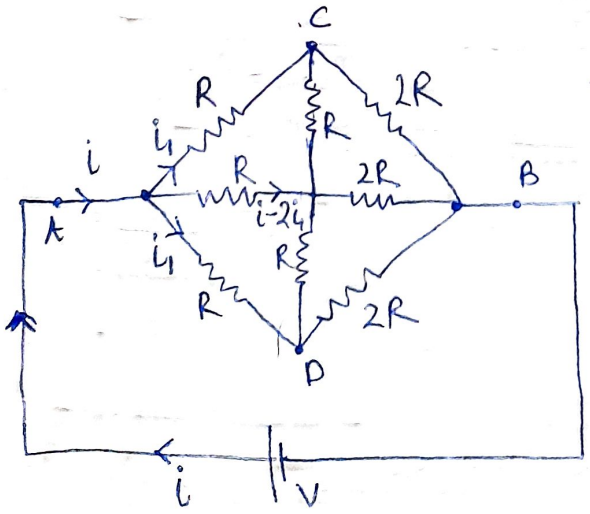
$$= R_{eq}^2 - R_{eq}R - 4R^2 = 0$$

$$R_{eq} = \frac{R \pm \sqrt{R^2 + 4 \times 4R^2}}{2}$$

$$R_{eq} = \frac{R + \sqrt{17}R}{2}$$

Equivalance Resistance of Symmetric circuits :

Q

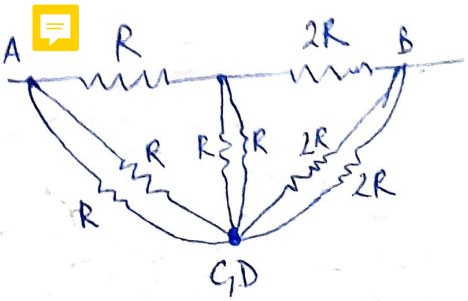


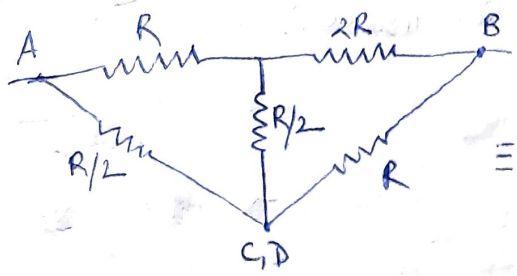
Method 1: Find R_{eq} across AB

$V_C = V_A - i_1 R = V_D$

$\therefore V_C = V_D$

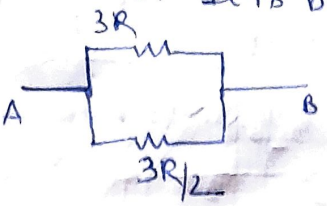
$\equiv$ Flip about AB





$R \times R = 2R \times \frac{R}{2} \Rightarrow R^2 = R^2$

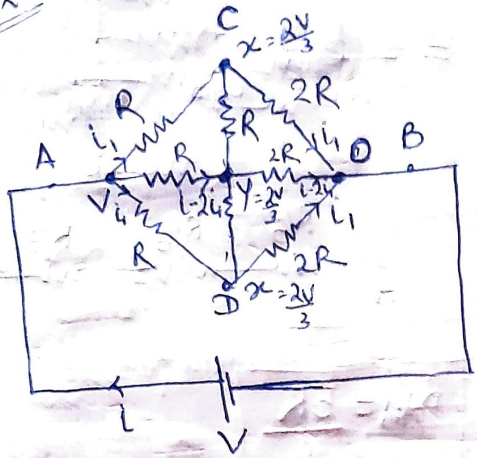
$\therefore$ It is balanced W.S bridge



$R_{eq} = \frac{3R \times 3R}{3R + 3R}$

$\therefore \boxed{R_{eq} = R}$

Method 2:



Using KCL at x:

$$\frac{x-V}{R} + \frac{x-y}{R} + \frac{x}{2R} = 0$$

$$2x - 2V + 2x - 2y + x = 0$$

$$5x - 2y = 2V \quad -(1)$$

Using KCL at y:

$$\frac{y-V}{R} + \frac{y}{2R} + \frac{(y-x)}{R} \times 2 = 0$$

$$2y - 2V + y + 4y - 4x = 0$$

$$7y - 4x = 2V \quad -(2)$$

$(-4x + 7y = 2V) \times 5$

$+ \quad (5x - 2y = 2V) \times 4$

$35y - 8y = 10V + 8V$

$27y = 18V$

$\boxed{y = \frac{2V}{3}}$ $\boxed{x = \frac{2V}{3}}$

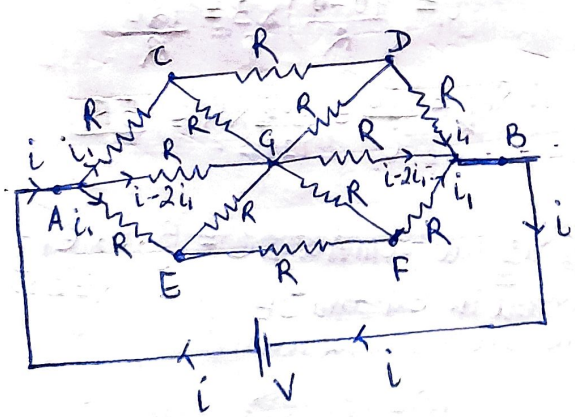
Now, $i = \frac{2V - 0}{3} \times 2 + \frac{2V - 0}{3}$ / $2R$

$\boxed{i = \frac{V}{R}}$

We know, By definition $R_{eq} = \frac{V}{i} = \frac{V}{V} \times R$

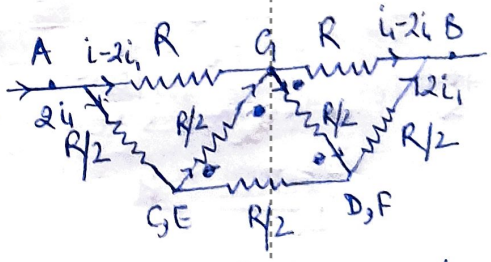
$\therefore \boxed{R_{eq} = R}$

Q

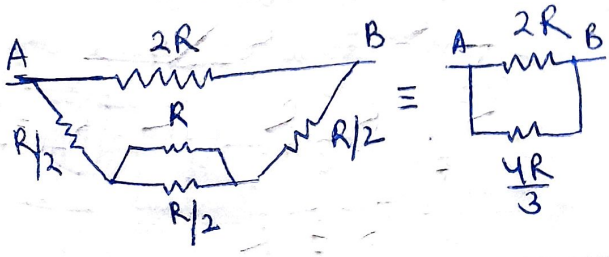


Find R_{eq} across AB.

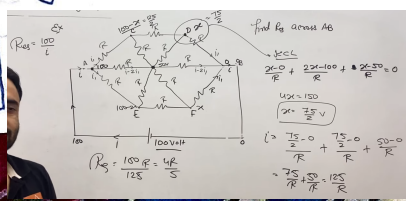
Flip ⟹



Current in C, E to G and G to D, F are same, they are in series.

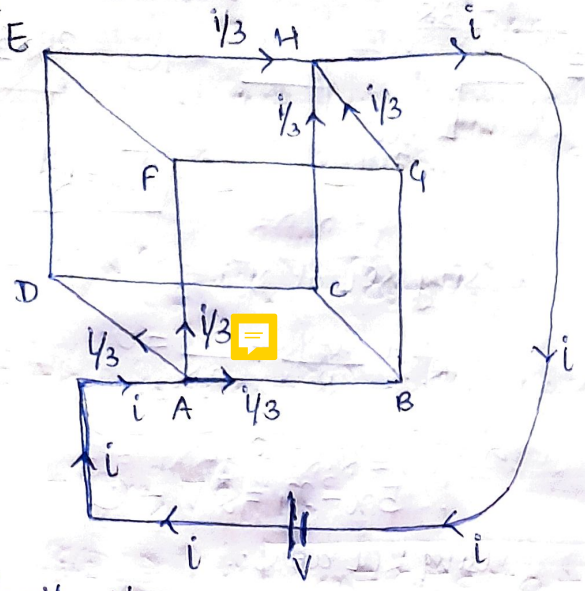


$\boxed{R_{AB} = \frac{4R}{5}}$

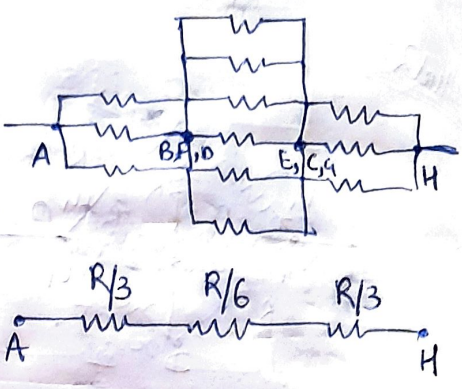


Q 12 resistor wire : each R Find Req across AH.

Method 1:



$\equiv$

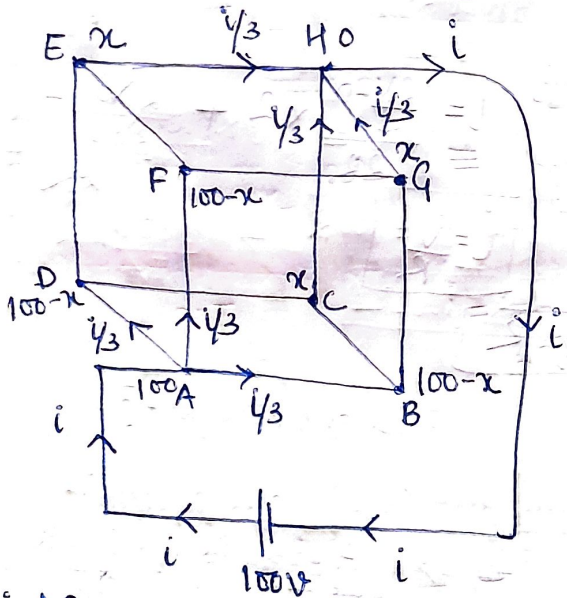


$\therefore$ ① $V_B = V_D = V_F$

② $V_E = V_G = V_C$

$$R_{AH} = \frac{5R}{6}$$

Method 2:



Using KCL at G:

$$\frac{x-0}{R} + \frac{2x-100}{R} + \frac{2x-100}{R} = 0$$

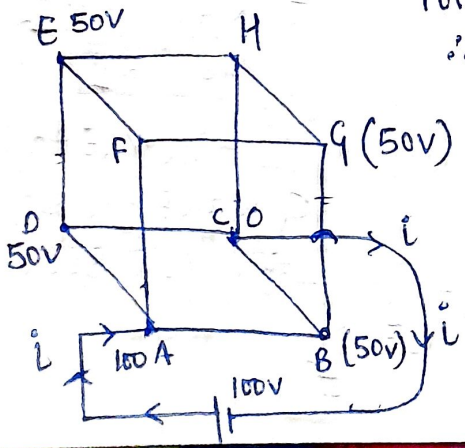
$$5x = 200$$

$$x = 40V$$

$$i = \left(\frac{40-0}{R}\right) \times 3 = \frac{120}{R}$$

$$\therefore R_{eq} = \frac{100}{i} = \frac{100}{120} \times R = \frac{5R}{6}$$

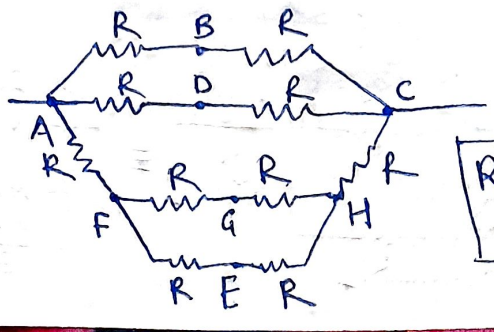
Q Find Req across AC.



Point equidistant from A and C = B, D, E, G (V = 50V)

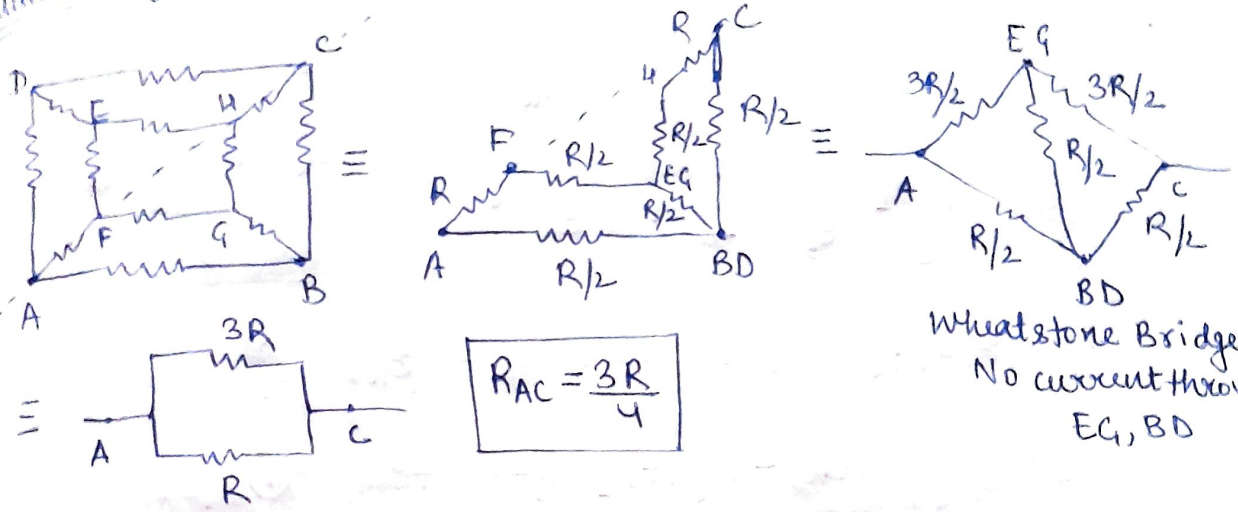
$\therefore$ No current in BG and DE so we can remove them.

$\equiv$



$$R_{AC} = \frac{3R}{4}$$

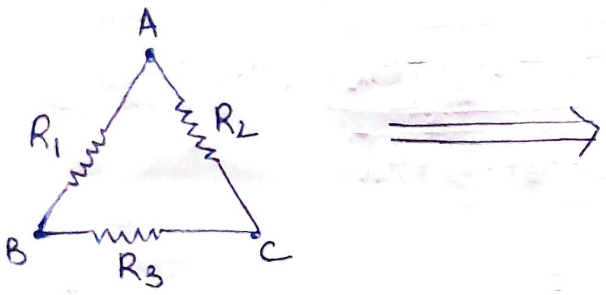
Method 2: Symmetry



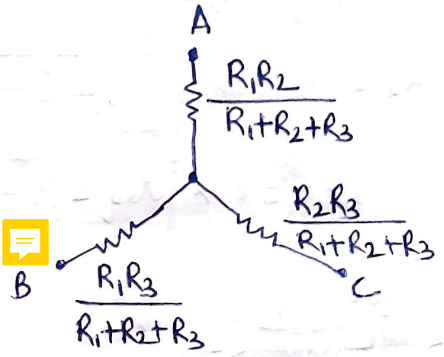
Wheatstone Bridge
No current through EG, BD

$$R_{AC} = \frac{3R}{4}$$

Delta to Star transformation:



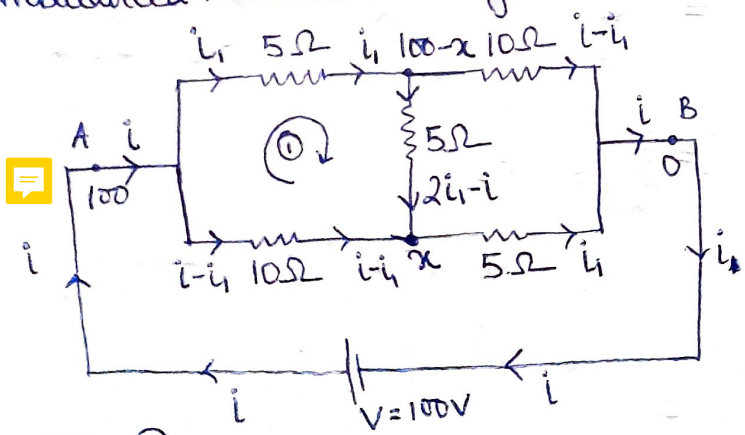
Delta Form



Star Form

Imbalanced Wheatstone Bridge:

Q



Find R_{eq} Across AB.

Method 1: Using KVL:

$$V - 5i_1 - 10(i - i_1) = 0$$

$$V - 5i_1 - 10i + 10i_1 = 0$$

$$V + 5i_1 - 10i = 0 \quad \text{—①}$$

Now, KVL in loop ①:

$$-5i_1 - 5(2i_1 - i) + 10(i - i_1) = 0$$

$$-i_1 - 2i_1 + i + 2i - 2i_1 = 0$$

$$5i_1 = 3i \quad \text{—②}$$

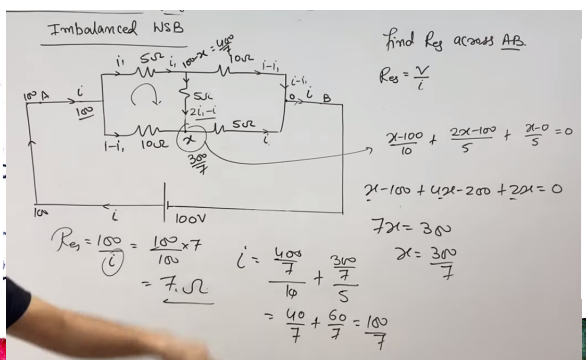
Put ② in ①

$$V = 7i$$

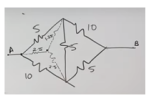
Now, $R_{eq} = \frac{V}{i} = \frac{7i}{i} = 7\Omega$

$$\therefore R_{eq} = 7\Omega$$

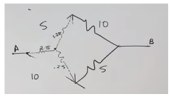
Method 2:

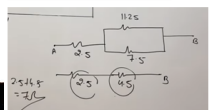


Method-3 : Star-Delta Transformation



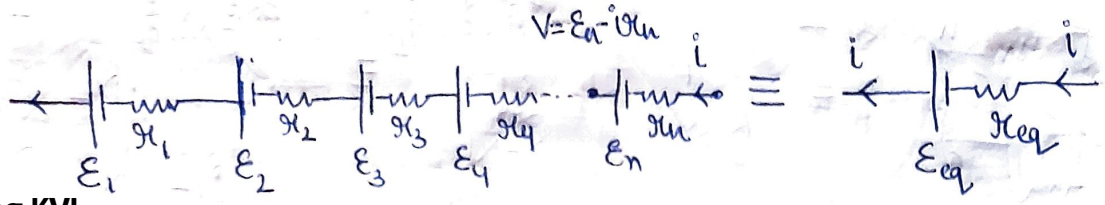
Two Deltas are present, convert any one into Star form; introduce a junction b/w the three junctions, write the resistances on those junctions are remove the existing Delta form resistances
now calculate the equivalent resistance





Grouping of Cells:

① Series:



applying KVL

$$(\varepsilon_1 - iR_1) + (\varepsilon_2 - iR_2) + (\varepsilon_3 - iR_3) + \cdots + (\varepsilon_n - iR_n)$$

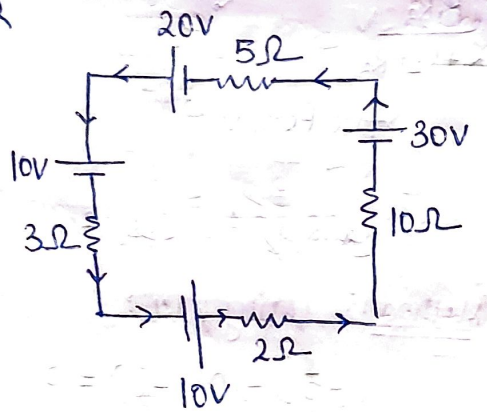
$$= (\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \cdots + \varepsilon_n) - i(R_1 + R_2 + R_3 + \cdots + R_n) \quad -①$$

$$\varepsilon_{eq} - iR_{eq} \quad -②$$

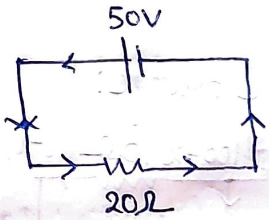
From ① and ② ⇒ ★ $\boxed{\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \cdots + \varepsilon_n}$ (with polarity)

★ $\boxed{R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n}$

Q



Find current in the circuit



$\varepsilon_{eq} = 30 + 20 - 10 + 10$

$\varepsilon_{eq} = 50\ V$

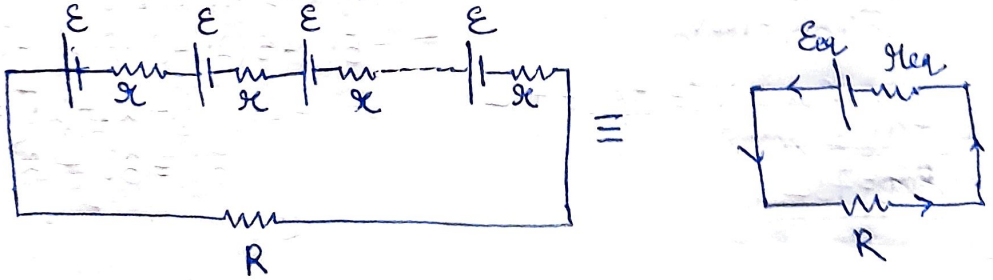
$R_{eq} = 5 + 3 + 10 + 2$

$= 20\ \Omega$

$$i = \frac{50}{20} = 2.5A$$

Special Cases:

① If n-identical cells (ε, r) are connected in series:

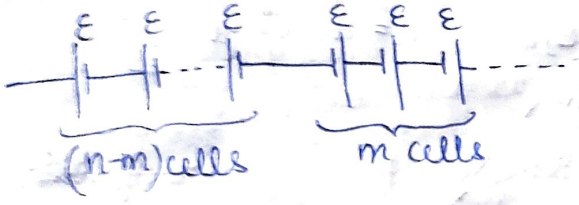


$$\boxed{i = \frac{\varepsilon_{eq}}{R + R_{eq}} = \frac{n\varepsilon}{R + nr}}$$

if $R >> nr$ → Condition for which series combination is advantageous to draw more current.

$$\boxed{i = \frac{n\varepsilon}{R}} \Rightarrow \boxed{i \propto n}$$

7) In n cells are in series out of which m cells are reversely connected



$$\mathcal{E}_{eq} = (n-m)\mathcal{E} - m\mathcal{E}$$

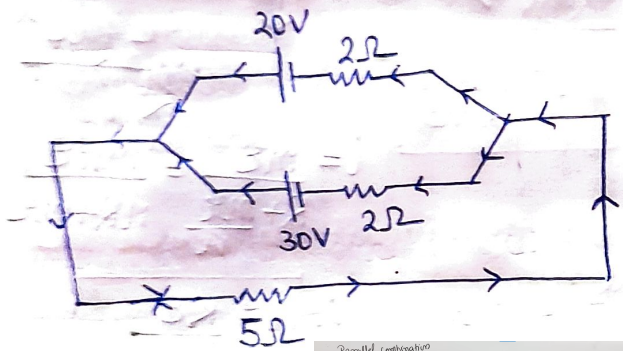
$$\boxed{\mathcal{E}_{eq} = (n-2m)\mathcal{E}}$$

8) Parallel Combination:

$$\boxed{\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}}$$

$$\boxed{\mathcal{E}_{eq} = \frac{\frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}} \quad \text{(with polarity)}$$

9)

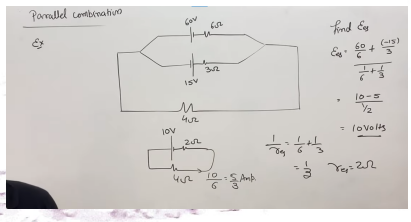


Find $\mathcal{E}_{eq}$ and current through 5Ω

$$\mathcal{E}_{eq} = \frac{\frac{20}{2} + \frac{30}{2}}{\frac{1}{2} + \frac{1}{2}} = 10 + 15 = 25V$$

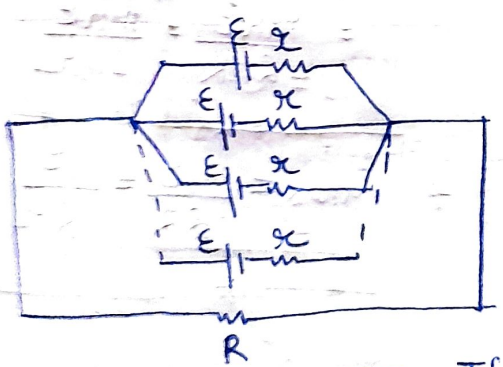
$$\frac{1}{r_{eq}} = \frac{1}{2} + \frac{1}{2} \Rightarrow \boxed{r_{eq} = 1\Omega}$$

$$\boxed{i = \frac{25}{6} A}$$

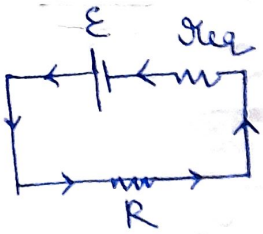


Special case:

If n-identical cells $(\mathcal{E}, r)$ are connected in parallel:



≡



$$\mathcal{E}_{eq} = \frac{\frac{n\mathcal{E}}{r}}{\frac{n}{r}} = \mathcal{E}$$

$$\boxed{r_{eq} = \frac{r}{n}}$$

$$\boxed{i = \frac{\mathcal{E}_{eq}}{R + r_{eq}} = \frac{\mathcal{E}}{R + \frac{r}{n}} = \frac{n\mathcal{E}}{nR + r}}$$

If $r >>> nR$ ⟶ Condition for which parallel combination is advantageous to draw more current

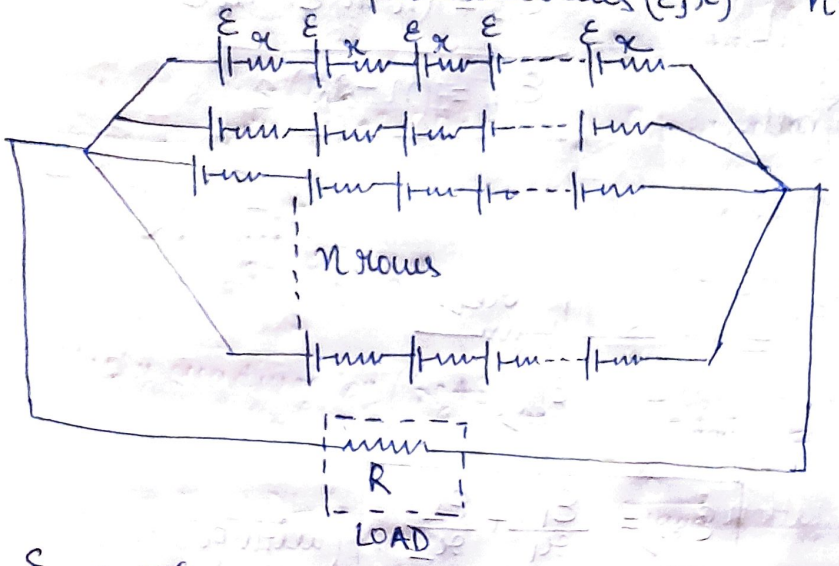
$$\boxed{i = \frac{n\mathcal{E}}{r}} \Rightarrow \boxed{i \propto n}$$

Grid of cells:

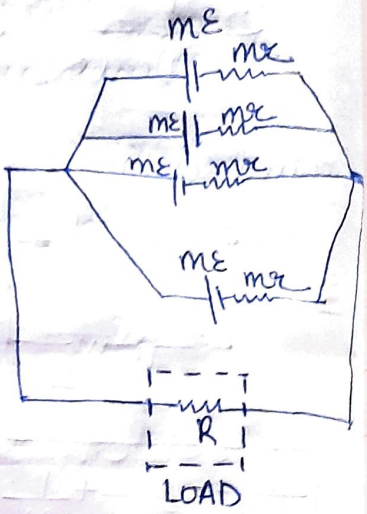
$R = mn$ = number of identical cells $(\mathcal{E}, r)$

m cells in a row

n = no. of rows.

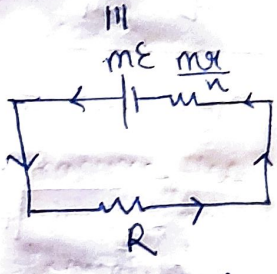


$\equiv$



$$\mathcal{E}_{eq} = \frac{\frac{m\mathcal{E}}{mr} + \frac{m\mathcal{E}}{mr} + \text{--- } n\text{ times}}{\frac{1}{mr} + \frac{1}{mr} + \text{--- } n\text{ times}} \quad \star \boxed{r_{eq} = \frac{mr}{n}}$$

$$= \frac{\frac{n\mathcal{E}}{r}}{\frac{n}{mr}} = m\mathcal{E} \quad \therefore \boxed{\mathcal{E}_{eq} = m\mathcal{E}}$$



$$i = \frac{m\mathcal{E}}{R + \frac{mr}{n}} = \frac{mn\mathcal{E}}{nR + mr} \quad \text{i)}$$

Now, $i = \frac{K\mathcal{E}}{nR + mr}$

$$\therefore \boxed{i = \frac{K\mathcal{E}}{nR + mr}}$$

For max^m i, $f(m,n) = nR + mr$ must be min^m

$$\Rightarrow f(n) = nR + \frac{K}{n} r$$

$$f'(n) = R + Kr\left(\frac{-1}{n^2}\right) = 0$$

$$= R - \frac{Kr}{n^2} = 0 \Rightarrow R = \frac{mnr}{n^2}$$

$$\therefore \boxed{nR = mr}$$

↳ Condition for maximum current.

$$i_{max} = \frac{mn\mathcal{E}}{nR + nR} = \frac{mn\mathcal{E}}{mr + mr}$$

$$= \frac{mn\mathcal{E}}{2nR} = \frac{mn\mathcal{E}}{2mr}$$

$$\star \boxed{i_{max} = \frac{m\mathcal{E}}{2R} = \frac{n\mathcal{E}}{2r}}$$

Instruments:

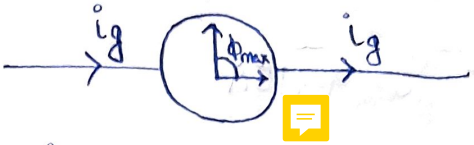
① Galvanometer / Micro-ammeter:

→ Used to detect current

→ It can measure current if current is very small.

Galvanometer

ϕ → deflection in needle



ig → Current for max^m deflection (usually in order of mA, μA).

$$\text{Current Sensitivity} = \frac{\phi}{i} = \frac{\phi_{max}}{i_g}$$

= deflection per unit current

★ Galvanometer can measure current if it is less than or equal to ig.

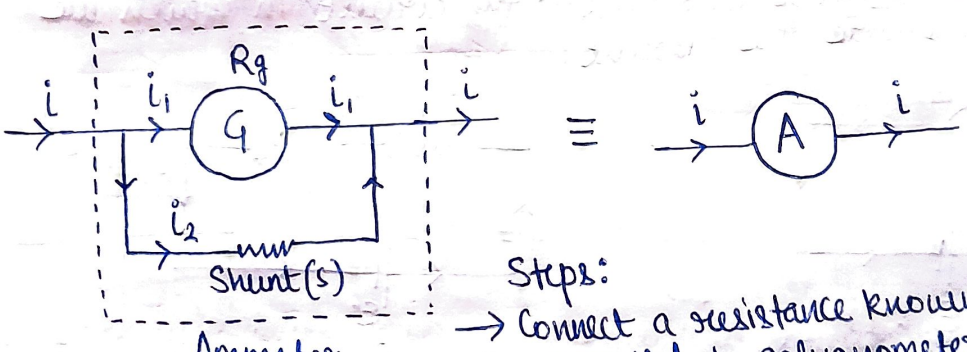
→ If it is said that galvanometer is damaged / faulty / harmed ⇒ $i > i_g$

② Ammeter:

→ It is modified galvanometer

→ It can measure $i > i_g$.

(i) How to make ammeter?



Ammeter

Steps:

→ Connect a resistance known as shunt (S), parallel to galvanometer.

→ Smaller the resistance of shunt, better it is. (So as to measure more current).

→ Calibrate the readings of galvanometer.

→ Multiply all the readings with $\left(\frac{R_g+S}{S}\right)$.

→ Maximum current ammeter can measure is $i_{max} = \left(\frac{R_g+S}{S}\right) i_g$

$$i_1 = \frac{S}{R_g+S} i$$

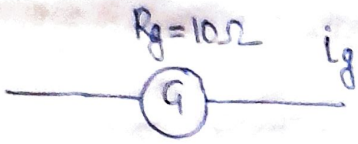
$$\therefore i = \left(\frac{R_g+S}{S}\right) i_1$$

Rg → Known

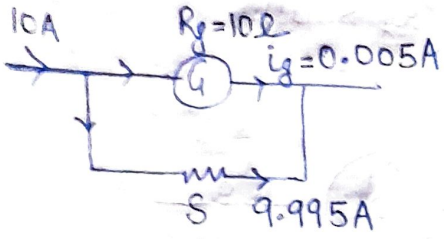
S → known

$\left(\frac{R_g+S}{S}\right)$ → Known

Q



What modifications are required to convert this galvanometer into an ammeter which can measure a current upto 10A?



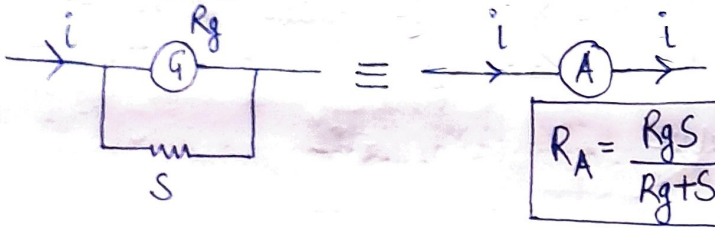
Potential difference across G and S is same.

$= 0.005 \times 10 = 9.995 \times S$

$S = \frac{5 \times 10}{9995} = \frac{10}{1999}\ \Omega$ $\therefore$ $\boxed{S = \frac{10}{1999}\ \Omega}$

Multiply readings by $\left(\frac{R_g + S}{S}\right) = \frac{10 + \frac{10}{1999}}{\frac{10}{1999}} = 2000$

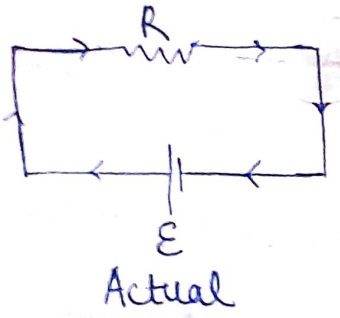
(ii) How to use ammeter?



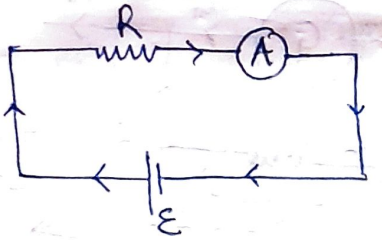
$\boxed{R_A = \frac{R_g S}{R_g + S}}$

Ammeter will measure the current passing through it.

* Connect the ammeter in series to the circuit in which we have to measure the current.



Actual



Experimental

$\boxed{i_{actual} = \frac{\varepsilon}{R}}$

$\boxed{i_{measured} = \frac{\varepsilon}{R + R_A}} < i_{actual}$

$\text{Error} = \frac{\varepsilon}{R + R_A} - \frac{\varepsilon}{R}$

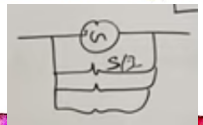
To minimize error, $R_A \downarrow$

$R_A = \frac{R_g S}{R_g + S}$ $\Rightarrow S \downarrow$

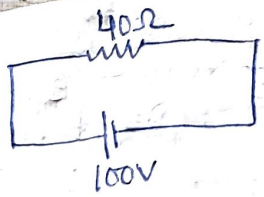
Ideal ammeter

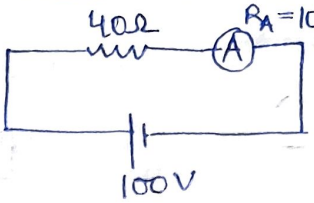
$\Rightarrow R_A \to 0$

$\Rightarrow S \to 0$



Q





① error = 2 − 2.5 = −0.5A

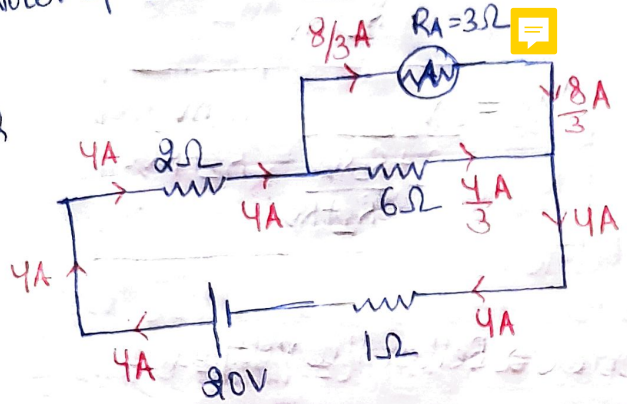
② fractional error = $\frac{-0.5}{2.5} = -0.2$

③ % error = −0.2 × 100 = −20%.

$i_{actual} = \frac{100}{40} = 2.5A$ $\quad i_{measured} = \frac{100}{50} = 2A$

Note: If it is said that ammeter is damaged ⇒ $\boxed{i > \left(\frac{R_g+S}{S}\right) i_g}$

Q



Find current through ammeter.

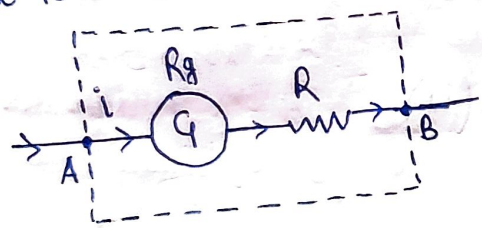
Current through (A):

$= \frac{R_{eq}}{R_A} i = \frac{2}{3} \times 4 = \frac{8}{3} A$

③ Voltmeter:

→ It is used to measure potential difference.

(i) How to make voltmeter?



≡

$R_V = R_g + R$

Voltmeter

$\boxed{V_A - V_B = i(R_g+R) = iR_V}$

i → current through galvanometer

$R_V = R_g + R$ (known)

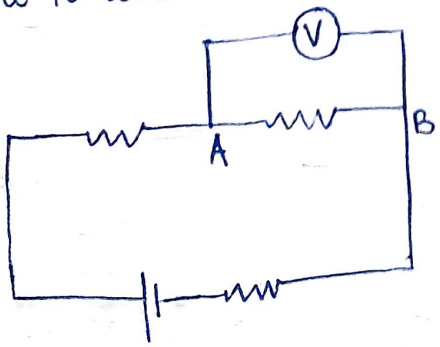
Steps:

→ Connect a high resistance R in series with galvanometer

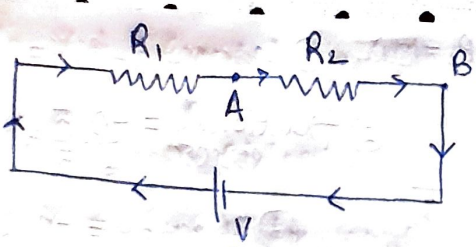
→ Calibrate the readings of galvanometer

→ Multiply all the readings with (R_g+R) and replace them

(ii) How to use voltmeter?



★ Connect a voltmeter parallelly b/w two points across whom we have to find potential difference.



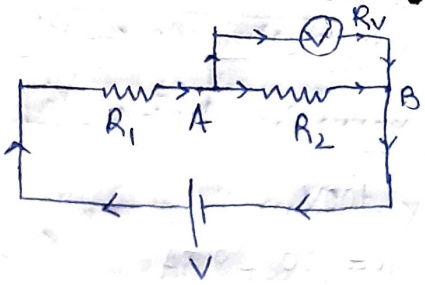
Actual

$$i = \frac{V}{R_1+R_2}$$

$$\boxed{V_{AB} = \frac{V}{R_1+R_2} R_2}$$

To minimize the error, $R_V \to \infty$

$$\boxed{V_{AB} = \frac{V R_2}{R_1+R_2}}$$



$$i = \frac{V}{R_1 + \frac{R_2 R_V}{R_2+R_V}}$$

$$\boxed{V_{AB} = \frac{V}{R_1+\frac{R_2R_V}{R_2+R_V}} \cdot \frac{R_2R_V}{R_2+R_V}}$$

when Rv tends to infinity, the measured P.D. tends to the actual P.D.

$\therefore$ Higher the $R_V \Rightarrow$ lesser will be error $\Rightarrow$ Better the voltmeter.

$\Rightarrow$ Ideal Voltmeter $\Rightarrow$ Error $\to 0 \Rightarrow R_V \to \infty \Rightarrow R \to \infty$

Q A galvanometer has 100 equal division. Its current sensitivity is $10 \frac{div}{mA}$. Its voltage sensitivity is $2 \frac{div}{mV}$.

① Find current for maximum deflection (i_g).

$C.S = \frac{\phi}{i}$

10 div $\to$ 1mA

1 div $\to \frac{1}{10}$ mA

100 div $\to \frac{1}{10} \times 100$ mA = 10mA

$\therefore \boxed{i_g = 10\,mA}$

② Find maximum potential difference across galvanometer.

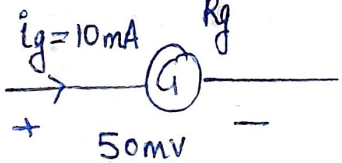
1mV $\to$ 2div

$\frac{1}{2}$mV $\leftarrow$ 1div

100 div $\to \frac{1}{2} \times 100$ mV = 50 mV

$\therefore \boxed{\text{Maxm p.d} = 50mV}$

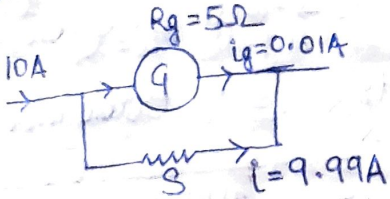
③ Find resistance of galvanometer.



$$R = \frac{V}{I} = \frac{50mV}{10mA} = 5\Omega$$

$\therefore \boxed{R_g = 5\Omega}$

④ What modifications are required to convert this galvanometer into ammeter which measure current upto 10A?



$\equiv$

② $R_A = \dfrac{5 \times \frac{5}{999}}{5 + \frac{5}{999}}$

$0.01 \times 5 = 9.99 \times S$

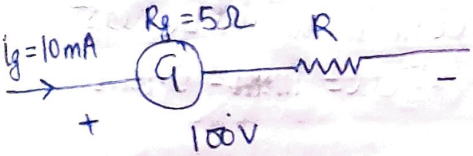
① $\boxed{S = \dfrac{5}{999}}$

③ Multiply readings with $\left(\dfrac{R_g + S}{S}\right)$

$= \dfrac{5 + \frac{5}{999}}{\frac{5}{999}} = 1000$

alternate : 10mA x multiplying factor = 1A

⑤ What modifications are required to convert this galvanometer into a voltmeter which can measure potential difference upto 100V?



$V = i_g (R_g + R)$

$100 = \dfrac{10}{1000}(5 + R)$

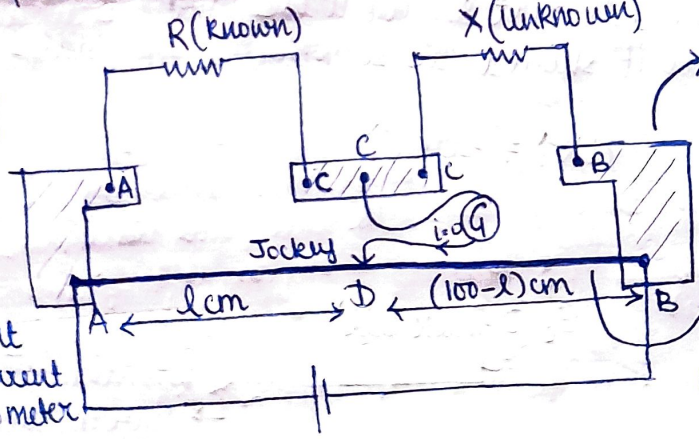
① $\boxed{R = 9995\Omega}$

② Multiply readings of galvanometer with $R_g + R = 10000\Omega$

Meter Bridge:

→ It is an experimental setup based on wheatstone bridge.

→ Used to find unknown resistance.

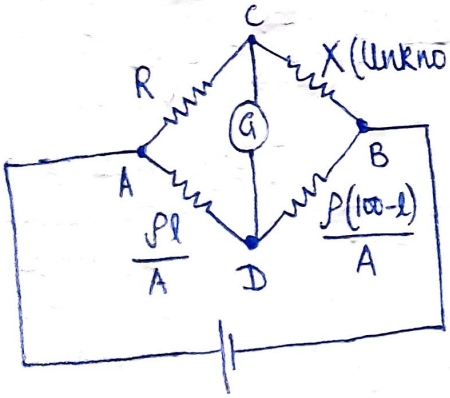


thus, there will be negligible potential drop across these plates

l → balancing length

D → Null point
↳ pt where current through galvanometer is zero.

$\equiv$



For no current through galvanometer:

$X \dfrac{\rho l}{A} = R \dfrac{\rho (100 - l)}{A}$

$\boxed{X = \dfrac{R(100 - l)}{l}}$

Now, $X = \frac{R(100-l)}{l}$

Take ln both sides

$\ln X = \ln R + \ln(100-l) - \ln l$

Differentiate both sides

$\frac{dX}{X} = \frac{dR}{R} + \frac{dl}{100-l} + \frac{dl}{l}$ { In error we always take mod }

$dX \rightarrow$ absolute error in X

$\frac{dX}{X} \rightarrow$ fractional error in X

$\frac{dX}{X} \times 100 =$ % error in X

Now, R = accurate $\Rightarrow dR = 0$

$\Rightarrow \frac{dX}{X} = \frac{dl}{100-l} + \frac{dl}{l} \Rightarrow \frac{dX}{X} = dl\left(\frac{l+100-l}{l(100-l)}\right) \Rightarrow \frac{dX}{X} = dl\left(\frac{100}{l(100-l)}\right)$

Now, $\frac{dX}{X} = \frac{dl \times 100}{l(100-l)}$

dl is absolute error in length.
Maxm possible error in length will be least count of scale. $\therefore dl = 1mm = 0.1cm$

$\Rightarrow \frac{dX}{X} = \frac{0.1 \times 100}{l(100-l)}$

Now, to minimise error, $dX \downarrow \Rightarrow$ Denominator must be maxm

$f(l) = 100l - l^2 \Rightarrow f'(l) = 100 - 2l = 0$

$\boxed{l = 50cm}$

★ To minimise error, null point should'nt lie near the end points of the wire, instead it should lie somewhere in middle of the wire.

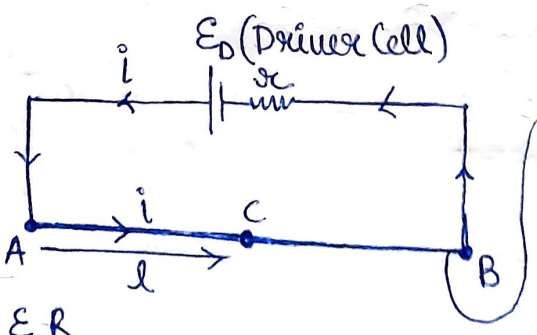
$l \rightarrow 0cm$
$l \rightarrow 100cm$ } $dX \uparrow \Rightarrow$ error $\uparrow$

$\Delta Q = mS\Delta T$

$i^2Rt = \Delta Q = C(\Delta T) \uparrow$

$l \uparrow \quad R \uparrow$

Potentiometer:



$i = \frac{\mathcal{E}}{R+r}$

$V_A - V_B = V = iR = \frac{\mathcal{E}R}{R+r}$

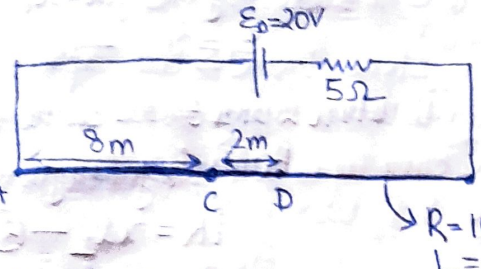
$\therefore$ Vmax that can be measured using potentiometer $= \frac{\mathcal{E}R}{R+r}$

★ Potential gradient $(x) = \frac{V}{L} = \frac{\mathcal{E}}{R+r}\frac{R}{L}$ (V/m) potential difference per unit length

Potentiometer wire:
→ Uniform wire of uniform cross-section (R, l)
→ High heat capacity
→ Melting point High.
→ Alloys used to make potentiometer wire: Nichrome, constantan, manganin.

* $V_A - V_C = xl = \frac{\mathcal{E}}{R+r} \cdot \frac{R}{L} l$

Ex.



① $i = \frac{20}{105} = \frac{4}{21}$ A

② $V_A - V_B = iR$
$= \frac{4}{21} \times 100 = \frac{400}{21}$ Volts

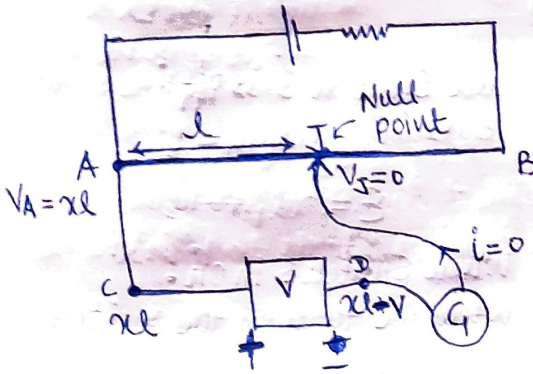
③ $V_{max} = \frac{400}{21}$ Volts

④ Potential gradient $(x) = \frac{400/21}{20} = \frac{20}{21}$ V/m

⑤ $V_A - V_C = xl = \frac{20}{21} \times 8 = \frac{160}{21}$ Volts

⑥ $V_C - V_D = xl = \frac{20}{21} \times 2 = \frac{40}{21}$ Volts

* How to use potentiometer to find potential difference?



Balancing length = l

So, for no current in galvanometer,

no current thru G
despite it having some resistance
hence, P.D. across it = 0

$xl - V = 0$

$\boxed{V = xl}$

OR we can say, at balancing length,

$V_A - V_J = V_C - V_D$

$xl = xl - xl + V$

$\boxed{V = xl}$

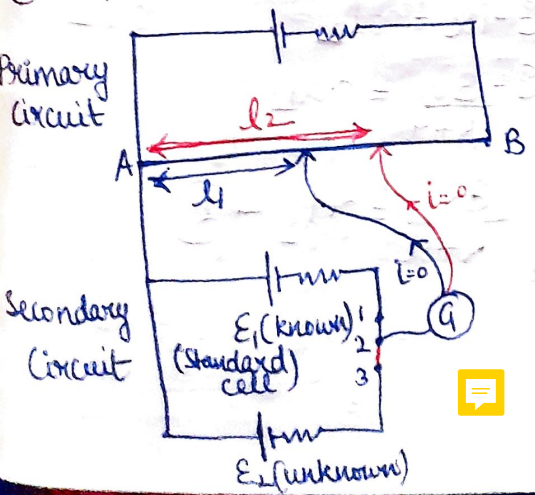
& what if there is no null points on potentiometer wire?

→ $V_{unknown}(V) > V_A - V_B$

→ lower polarity of $V_{unknown}$ is connected to point A.

Applications of potentiometer:

① To find unknown emf of a cell.



(i) When 1 and 2 are connected

Balancing length = l_1

$\mathcal{E}_1 = xl_1$ —— ①

(ii) When 3 and 2 are connected

Balancing length = l_2

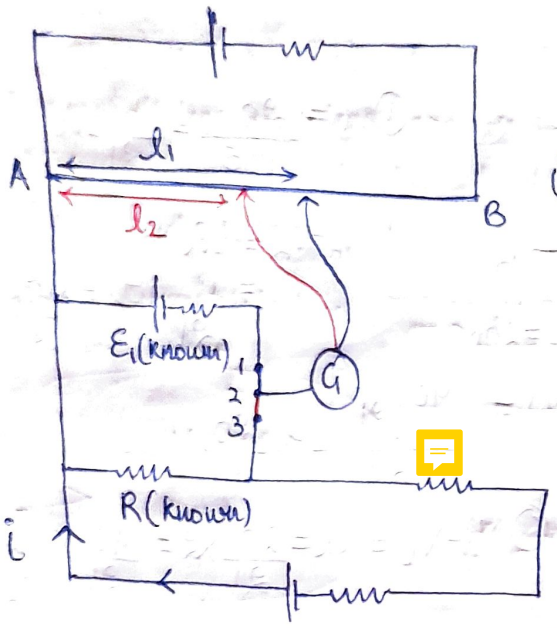
$\mathcal{E}_2 = xl_2$ —— ②

From ① and ②

$\frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{l_1}{l_2}$

$\therefore$ $\boxed{\mathcal{E}_2 = \frac{l_2}{l_1}\mathcal{E}_1}$

② To find current in a circuit.



(i) When 1 and 2 are connected.

Balancing length $= l_1$

$\mathcal{E}_1 = x l_1$ — ①

(ii) When 2 and 3 are connected.

Balancing length $= l_2$

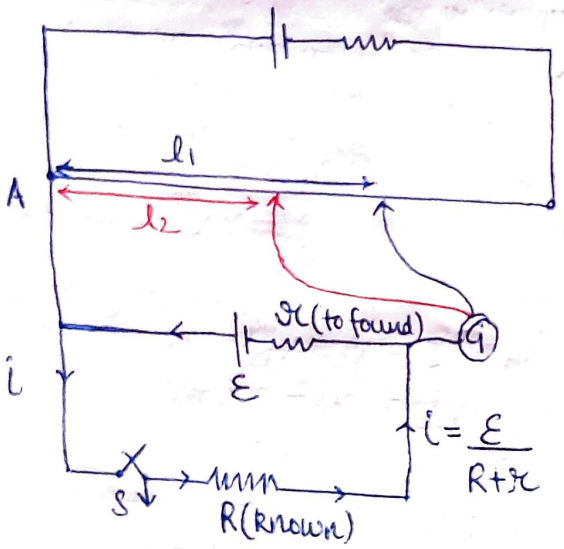
$iR = x l_2$ — ②

From ① and ②

$$\frac{iR}{\mathcal{E}_1} = \frac{l_2}{l_1}$$

$$\star \quad \boxed{i = \frac{\mathcal{E}_1}{R}\frac{l_2}{l_1}}$$

③ To find internal resistance of a battery.



(i) When Switch S is off:

Balancing length $= l_1$

$\mathcal{E} = x l_1$ — ①

(ii) When Switch S is on:

When we turn on the switch, current flows in secondary circuit. Potential difference across cell now decreases to $\mathcal{E} - ir$. So the null point will now be found only on the left of l_1.

Balancing length $= l_2$

$l_2 < l_1$

$\mathcal{E} - ir = x l_2$

$= \mathcal{E} - \frac{\mathcal{E} r}{R+r} = x l_2$

$= \frac{\mathcal{E} R}{R+r} = x l_2$ — ②

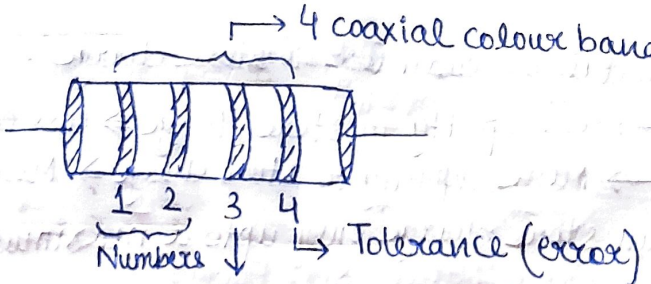
Divide ② and ①

$$\frac{R}{R+r} = \frac{l_2}{l_1}$$

$\Rightarrow R l_1 = R l_2 + r l_2$

$$\star \quad \boxed{r = \frac{R(l_1 - l_2)}{l_2}}$$

Colour Coding of Carbon Resistors:



* $R = N_1 N_2 \times 10^{N_3} \pm N_4\%$

Colour	Number (1,2)	Multiplier	Tolerance
Black	0	10^0	Gold = 5%
Brown	1	10^1	Silver = 10%
Red	2	10^2	No colour = 20%
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Grey	8	10^8	
White	9	10^9	

BBROY of Great Britain has a Very Good Wife

Q

Find Resistance.

$R = 16 \times 10^3 \pm 10\%$

Q

Find Resistance

$R = 27 \times 10^4 \pm 5\%$